



Large-scale optimization of nonconvex MINLP refinery scheduling

Robert E. Franzoi^{a,b}, Brenno C. Menezes^c, Jeffrey D. Kelly^d, Jorge A.W. Gut^{b,*},
Ignacio E. Grossmann^e

^a Andlinger Center for Energy and the Environment, Princeton University, Princeton, NJ, United States

^b Department of Chemical Engineering, Universidade de São Paulo, São Paulo, Brazil

^c Division of Engineering Management and Decision Sciences, College of Science and Engineering, Hamad Bin Khalifa University, Qatar Foundation, Doha, Qatar

^d Industrial Algorithms Limited, 15 St. Andrews Road, Toronto, Canada

^e Department of Chemical Engineering, Carnegie Mellon University, Pittsburgh, United States

ARTICLE INFO

Keywords:

Mathematical programming
Optimization
Refinery scheduling
Nonconvex MINLP
Decision-making framework
Heuristic algorithms

ABSTRACT

Modeling and optimization of large-scale refinery scheduling problems is challenging because of their complexity and size. Herein, we propose a mathematical model to represent such problems more accurately and realistically, and a state-of-the-art optimization framework for its solution. The framework leverages the use of mathematical optimization and algorithmic methods by combining modeling approaches (process design, model decompositions), solving strategies (rescheduling, heuristics), and machine learning regression (surrogate models). An industrial-size refinery scheduling problem (2 blenders, 4 feed tanks, distillation network with 5 towers, processing network with FCC, hydrotreaters, debutanizers, superfractionator, catalytic reformer) is formulated as a hierarchical nonconvex mixed-integer nonlinear programming (MINLP) model and is successfully optimized, providing higher profitability and more efficient scheduling operations considering 12 feedstocks, 10 products and multiple scenarios for time horizon and step. Results highlight the importance of tuning scheduling parameters and employing an enhanced computer-aided framework to enable the solution of industrial refinery scheduling operations.

1. Introduction

Over the last decades, crude oil refinery operations have grown increasingly complex. Factors including tighter competition, stricter environmental regulations, and lower profit margins, require companies to employ enhanced computer-aided decision-making capabilities for achieving higher efficiency and reducing costs in planning, scheduling, coordinating and supply chain applications (Shah et al., 2015). However, such operations are very complex and involve decision-making of a large number of feedstocks, flows, unit-operations, properties, and products within an enterprise-wide optimization (EWO) problem.

A key goal of crude oil refineries is to properly determine the scheduling operations, which comprise the decision-making of tasks to be performed in the plant, and significantly affect the economics and operations in the oil refineries. Scheduling operations are modeled as large-scale nonconvex mixed-integer nonlinear programming (MINLP) problems, which are challenging because of their complexity (combinatorial decisions, nonlinear and nonconvex equations) and size (large

number of variables and constraints). This imposes difficulties for the solution of complex industrial cases, which are challenging given the current state-of-the-art modeling tools and optimization capabilities (e. g., computing power, optimization solvers).

Several strategies have aimed to provide tractable applications for large-scale refinery operations. Recent methods include tight relaxations with piecewise McCormick or normalized multiparametric disaggregation (Castillo Castillo et al., 2017), cluster decomposition (Uribe-Rodriguez et al., 2020), and Lagrangean decomposition (Uribe-Rodriguez et al., 2023), although being more focused on a planning perspective.

Methods for large-scale refinery scheduling often introduce simplifications for the modeling of processing units, representation of the process network scope, and limiting search space in the optimization algorithms. However, not only are these simplifications often too drastic and unrealistic relative to the industrial needs, but there is still a huge gap between the state-of-the-art solution methods and the refinery scheduling problems that need to be solved in the industry. There have been recent advances in computer-aided resources, solution algorithms,

* Corresponding author.

E-mail address: jorgewgut@usp.br (J.A.W. Gut).

<https://doi.org/10.1016/j.compchemeng.2024.108678>

Received 18 December 2023; Received in revised form 8 March 2024; Accepted 29 March 2024

Available online 30 March 2024

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and decision-making (Franzoi et al., 2018a; Rafiei and Ricardez-Sandoval, 2020). Improved modeling, solving, implementation, and integration approaches, as well as machine learning and big data strategies, increasingly lead to opportunities to lower costs and improved operations. Commercial optimization solvers have become increasingly efficient, in addition to the enhancements in computational processing. Remarkable success has been achieved in improving refinery scheduling optimization, including the development of novel methods, models, and algorithms (Li et al., 2020). The remaining challenges rely on developing mathematical models and decision-making frameworks to address the typical industrial challenges and provide implementable solutions for industrial scheduling decision-making.

1.1. Current limitations on refinery scheduling decision-making

Simplifications have been traditionally introduced in refinery scheduling problems to achieve tractability. However, they often compromise the solution quality or lead to schedules that are not easily implementable in the plant. In the following, we highlight key considerations for the modeling and optimization of industrial-scale refinery scheduling.

First, due to modeling, solving, and computational limitations, building unit-operation models based on first principles and optimizing the integrated refinery units in multilevel decisions remains intractable. Hence, the refinery scheduling problem has been traditionally decomposed into smaller sub-problems that are modeled and solved separately, namely, crude oil or feed blending network, production and process network, and product blending network (Shah et al., 2015). Solving an integrated scheduling formulation comprised of all three segments is denoted herein as refinery scheduling. It is well established that better solutions are obtained by solving integrated formulations rather than their hierarchical (decentralized and/or distributed) counterparts (Yang et al., 2020). Feasible solutions found in the hierarchical approach may not be feasible in the integrated formulation, and the conditions for optimality are also different. Despite extensive literature on scheduling optimization for each refinery sub-problem, such an integrated scheduling formulation for industrial-sized applications has not been studied (Li et al., 2020). However, it should be highlighted that this is not necessarily due to the aforementioned technological issues but are also related to the current business, organizational and risk management hierarchical decompositions found in the refinery as well as the skill-level and experience of the personnel.

Second, standard refinery scheduling formulations do not include some of the key variables and requirements from the plant. Hence, their mathematical representations are not coherent with real-world industrial operations that involve a systematic production with detailed logistics and quality operations (Mouret et al., 2009; Castro and Grossmann, 2014; Xu et al., 2017). As pointed out by Franzoi et al. (2019), it is imperative to account for a detailed and accurate refinery network. Moreover, it is fundamental to introduce constraints that represent logistics limitations, such as drawing and filling operations in tanks (e.g., to avoid simultaneous operations), operational modes (i.e., same physical unit operating with different feeds and products at distinct conditions), minimum unit operating time, and unit downtimes (i.e., shut-down or maintenance periods). These are useful to incorporate relevant plant requirements into the mathematical model (Zyngier and Kelly, 2009). The lack of such constraints in refinery scheduling formulations compromises the implementation of the solution in the plant.

Third, several industrial processes still employ trial-and-error decision-making i.e., scheduling-by-simulation, rather than optimization-based approaches (Chrysosoulouris et al., 2005; Zhang et al., 2010). However, the search space and exploration employed by optimization methods are greatly superior and especially important for industrial operations, which comprise complex, uncertain, and dynamic processes and that consider process safety, environmental impact, operational

requirements, scheduling implementation, and plant-model mismatches. Employing optimization approaches for scheduling problems provides several benefits in terms of improved operating performance and profitability.

Fourth, first principles formulations are often employed for processing units throughout the refinery, such as distillation towers, reactors and blenders. However, they are computationally expensive and not suitable for scheduling optimization. Instead of using complex or rigorous models to model these units, simplified empirical-based and data-driven correlations may be employed (e.g., surrogate equations), which allow the integration of processing unit models in refinery scheduling optimization applications (Cuadros Bohorques et al., 2020; Franzoi et al., 2020). These surrogates are supposedly accurate representations that provide relevant additional information at a low computational cost.

Fifth, proper calibration of scheduling parameters such as time-step and time horizon is fundamental. On one hand, the computational effort is key and imposes limitations on the model complexity. On the other hand, an ideal solution would benefit from the gains achieved with small time-steps and a large time horizon. Small time-steps may provide better decision-making and better solutions due to the additional degrees of freedom available, which results in better management of resources and production configuration. A long time horizon provides additional information that impacts the optimal scheduling through spot market opportunities and more efficient production and management of resources at the risk of dealing with increased uncertainty.

Sixth, noises, disturbances, disruptions, distractions and unforeseen events happen over time and lead to plant-model mismatches, which significantly worsen operations and economics. Some optimization fields such as stochastic programming and robust optimization aim to provide proper methodologies for such problems. Although efficient for diverse applications, stochastic approaches require scenario-based probability functions that may be difficult to estimate or that needlessly increase the model complexity, and may fail to handle constrained problems because the stochastic search operators often produce infeasible solutions (Francisco et al., 2005); and robust optimization methods are very conservative as they aim to hedge against all possible worst-case uncertain scenarios, which often leads to overly expensive or needlessly complex solutions (Garcia et al., 2016). Therefore, these approaches may not be the most suitable alternatives for refinery scheduling. Conversely, re-optimization procedures, known as online scheduling or rescheduling, provide efficient capabilities for revising the optimal schedule whenever necessary or beneficial, thereby mitigating plant-model mismatches (Franzoi et al., 2021a). This is especially important in complex processing sites that require integrated plant-model environments for data updating and accurate tracking of quality information throughout the process network (Franzoi et al., 2018a).

1.2. Objectives and contributions of this work

The objective of this paper is threefold. First, we discuss the state-of-the-art limitations of large-scale refinery scheduling applications, and we provide an overview of efficient modeling and optimization methods for refinery scheduling. Second, we present a novel mathematical model that comprises process-unit models and operational constraints particularly relevant to industrial scheduling operations. Third, we propose a novel decision-making framework and illustrate its application to solve an industrial-scale nonconvex MINLP refinery scheduling problem.

There are six particularly important features addressed herein, some of which have not been typically considered in literature and industry. From a modeling perspective, this includes the consideration of an integrated refinery scheduling network, coherent mathematical formulation based on plant requirements, and advanced process-unit modeling. Specifically, we propose a novel mathematical model that represents refinery scheduling operations in a more accurate and realistic fashion.

The model comprises an integrated refinery topology with crude oil blending network, production network, and product blending network; critical constraints are introduced to manage the logistics operations of units and tanks; and the process-unit models for the crude oil distillation towers are represented through data-driven surrogate equations for improved accuracy.

From a solving perspective, this includes optimization decision-making, consideration of scheduling parameters, and re-optimization mechanisms. We propose a novel decision-making framework that employs advanced modeling and solving methods, and which relies on optimization decision-making. We study and evaluate how scheduling parameters affect the solution method. Finally, we indicate how our proposed methodology is suitable for re-optimization mechanisms within a moving horizon fashion, which aims to further improve the scheduling solution and provide desired characteristics for industrial applications.

The framework is shown to efficiently solve large-scale nonconvex MINLP refinery scheduling problems comprised of hundreds of thousands of variables and constraints that to the best of our knowledge have never been addressed in such complexity in the literature. It comprises a) discrete-time modeling using the unit-operation-port-state superstructure (UOPSS) representation; b) temporal decomposition; c) phenomenological decomposition of the quantity, logic and quality dimensions; d) efficient design for the mathematical formulation (model design) and the operations (process design); e) model reduction strategies; f) relaxation heuristics for MILP models; g) rescheduling strategies to minimize plant-model mismatches by handling process uncertainties and disturbances; h) surrogate models as an alternative to their rigorous counterparts.

This paper is outlined as follows. An overview of refinery scheduling and algorithmic methods for refinery scheduling optimization is provided in Section 2. A novel mathematical formulation for refinery scheduling is presented in Section 3. A modeling and optimization decision-making framework is proposed in Section 4, which solves a large-scale refinery scheduling example presented in Section 5. The remarks and future outlook are highlighted in Section 6.

2. Literature review

2.1. Crude oil refinery network

The refinery network shown in Fig. 1 includes three segments, namely, crude oil or feed blend scheduling, production network, and product blending network.

The crude oil blend scheduling comprises the unloading and mixing operations of crude oil feedstocks. (Lee et al., 1996; Kelly and Mann,

2003; Saharidis et al., 2009; Castro and Grossmann, 2014). The crude oil blends are physically separated into distillates which are further converted into more valuable products. Because of the complexity of conversion operations, simplified correlations are often adopted in scheduling for refinery unit-operations, including distillation (Franzoi et al., 2020), fluid catalytic cracking (Cuadros Bohorquez et al., 2020), catalytic reforming (Mencarelli et al., 2020), hydrocracking (Song et al., 2021), hydrotreating (Xia et al., 2021), among others. The product blending segment concerns the mixing of intermediate refinery rundown streams into final products to be sold and distributed via logistics operations (Al-qahtani and Elkamel, 2011; Ali et al., 2022).

2.2. Algorithmic methods for industrial refinery scheduling optimization

The detailed mathematical formulation needed to represent refinery scheduling problems is difficult to solve as full-space nonconvex MINLP problems. To overcome the computational limitations that arise from the high complexity and combinatorial size of such systems, algorithmic methods based on heuristics and decompositions have been often employed. They provide significant benefits and have good cost-effectiveness in reducing computational effort while maintaining solution quality. In Supplementary Material A, we introduce strategies that are particularly useful for tackling large-scale refinery scheduling models. This includes the phenomenological decomposition heuristic, a linear programming reformulation for nonconvex equations, temporal decomposition, relaxation heuristic, surrogate modeling, and online optimization methods.

3. Mathematical formulation for refinery scheduling optimization

We propose a novel mathematical formulation for industrial-scale refinery scheduling optimization problems. The model is built based on discrete-time and uses the UOPSS representation. Previous works on the topic have shown good performance of the UOPSS representation for modeling and optimizing large-scale refinery scheduling (Kelly et al., 2017; Brunaud et al., 2020). Our proposed full-space MINLP model is comprised of three parts:

- The base formulation includes constraints to manage the relationships among unit-operations and port-states and to establish material balances throughout the network (Eqs. (1) to (15)). All constraints from the base formulation are also used in the MILP and NLP formulations.
- The MILP formulation (Logistics problem) includes the base formulation in addition to logistics constraints (Eqs. (16) to (37)), which

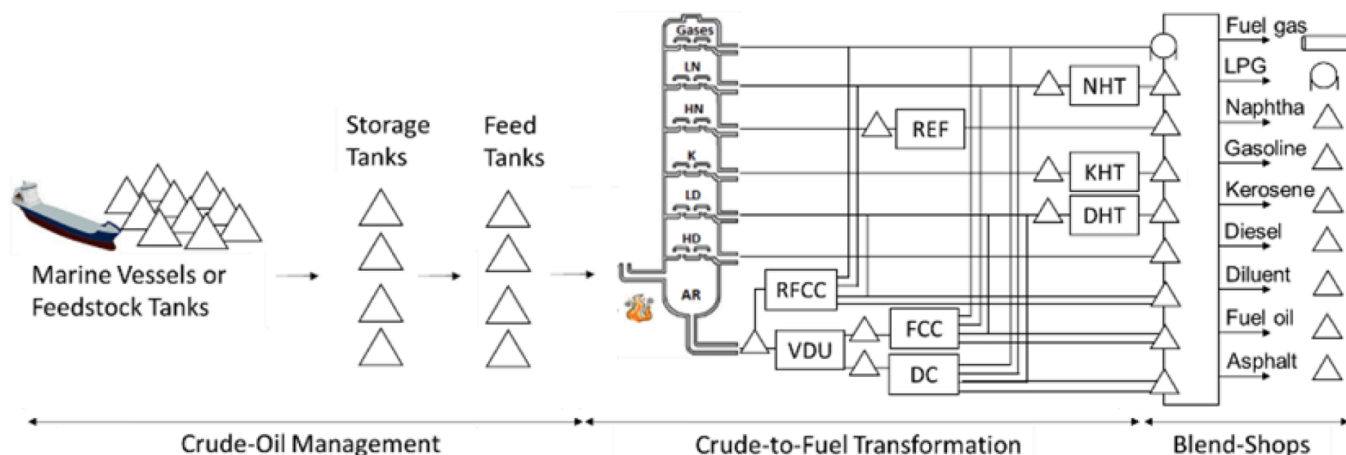


Fig. 1. Crude oil refinery network (adapted from Kelly et al., 2017).

involve: a) structural transitions and selection of operational modes; b) temporal transitions of unit-operations in sequence-dependent cycles; c) sharing of objects (units, ports, etc.) in multi-use constraints; d) operational time, uptime or runlength and zero downtime of units; e) fill-draw delay, fill-to-full, and draw-to-empty operations for tanks; and f) linear reformulation of blending equations.

- The NLP formulation (Quality problem) includes the base formulation in addition to the quality balances presented in Eqs. (38) to (43) for the blending constraints and Eqs. (44) to (46) for the material balances and transformations of crude oil components c into subsequent fractions. The NLP formulation cannot handle binary variables, so the UOPSS equations are reformulated either by setting the binary variables to unity or to their optimal values from previous logistics solutions.

The novelty of our formulation is twofold. First, we propose an **integrated** model for the entire refinery network, which includes all three segments (crude oil blend scheduling, production network, and product blending network) within an industrial-scale formulation. We introduce equations to estimate the outputs from process unit-operations based on **surrogate modeling** (for the **distillation** network), although models with **fixed yields and properties** are considered (for the other process units in the production network).

Second, we introduce critical constraints to represent the **logistics operations** in the refinery site. Although refinery scheduling operations are associated with many types of logistics limitations and requirements, mathematical formulations in the industry and literature typically neglect these aspects due to their high complexity, detailed information, and difficult implementation. The operational limitations introduced in

our model are fundamental to achieving realistic scheduling solutions. This is particularly relevant from an implementation perspective. Given that the logistics requirements are a critical part of refinery scheduling optimization, our novel formulation leads to solutions that can be more easily and smoothly implemented in the plant. Another advantage is that our constraints implicitly account for **feed changeover operations** in the distillation network. This eliminates the need of explicitly adding changeover equations, whose parameters are difficult to estimate and tune, leading to inaccuracies.

In this section, we introduce:

- Most important UOPSS characteristics, objects, and groups of objects.
- Lists of indices, sets, subscripts, superscripts, parameters, and variables.
- Equations, constraints, and objective function.

3.1. UOPSS objects

The network shown in Fig. 2 is comprised of a group of main objects that represent the sources and sinks ($\diamond$), tanks and inventories (Δ), and continuous processes ($\boxtimes$). These objects are referred to as unit-operations hereinafter. The unit-operations can be classified into four categories, namely, **perimeter-based**, **pool-based**, **mixer-based**, and **splitter-based**. There is a relationship between how the unit-operations of each category are represented and modeled in the formulation and their logistics and processing operations in the refinery site.

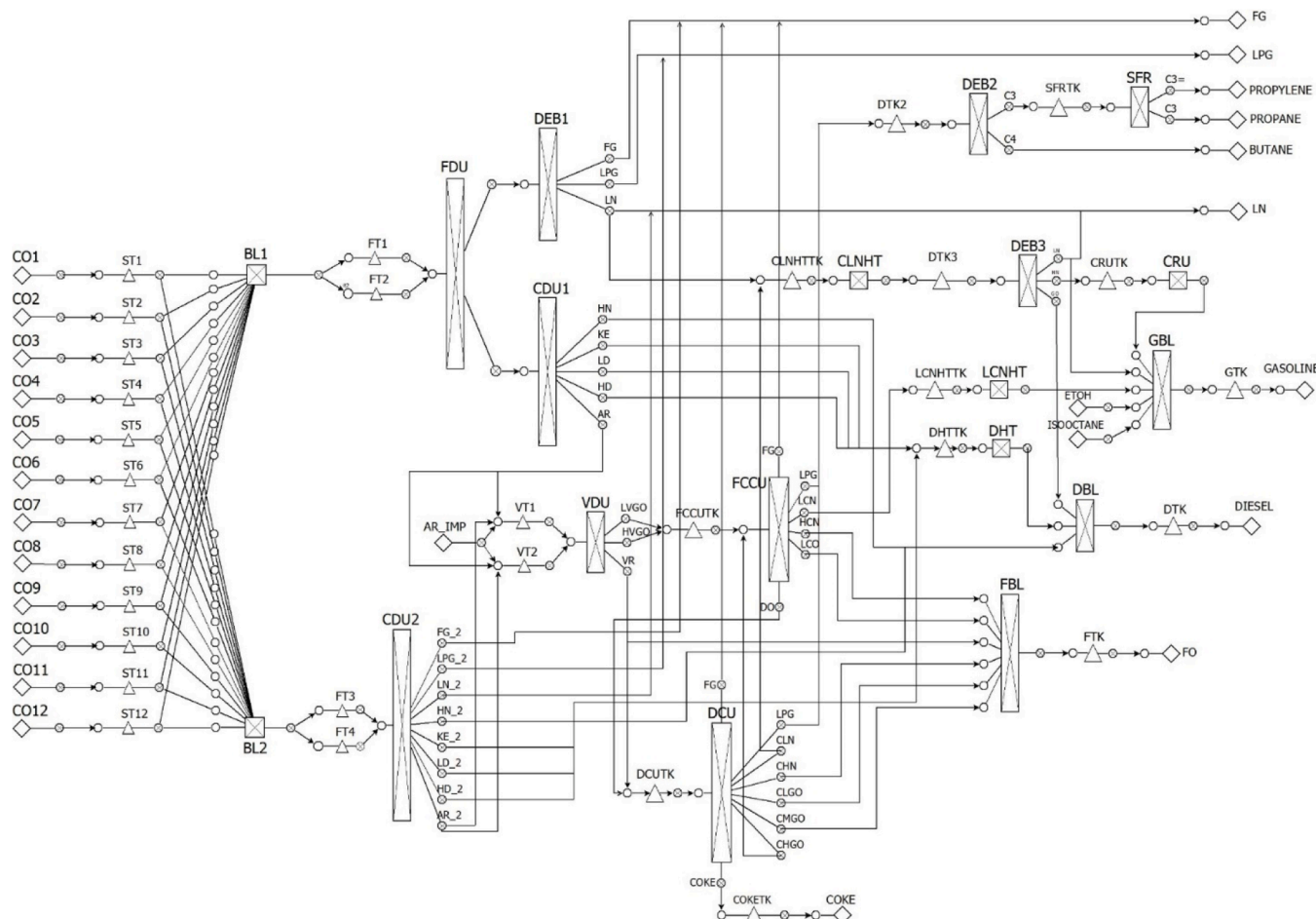


Fig. 2. Refinery scheduling optimization network.

- Perimeter-based (e.g., CO1) represent quantity-flows incoming to or outgoing from the plant (e.g., arrival of crude oils or delivery of final products through pipelines).
- Pool-based (e.g., ST1) refers to intermediate tanks throughout the plant, often allocated before or after processing units. They are important for ensuring smooth operations (e.g., constant feed for downstream units) and to provide logistics flexibility.
- Mixer-based (e.g., BL1) are continuous stirred blender units typically used to prepare the feed for processing units or for final product specifications by blending multiple intermediate streams with boosters (e.g., for the production of fuels). They can receive multiple simultaneous inlet streams, although a single outlet stream is operationally allowed.
- Splitter-based (e.g., CDU1) refers to physical and chemical processing units (e.g., distillation, fluid catalytic cracking, hydrotreating, delayed coking). For efficient operations and control, the profiles of the feed amount and quality should be as smooth as possible, with minimum or no variations over time. Splitter-based units support a single inlet stream, but can have multiple outlet streams, typically associated with the conversion and separation processes.

There are auxiliary objects before and after each unit-operation, referred to as in-ports and out-ports (O and $\otimes$) and arrows ($\rightarrow$) that represent connections between ports (e.g., material flows). The unit-operations are configurable to have **operational modes** (not shown in Fig. 2 for the sake of simplicity) and are connected to inlet and outlet ports. Ports are combined in pairs (j, i) to represent transfer or movement operations, whereby establishing material balances for each unit-operation.

States are used for configuring non-material flows and elements related to non-material flows (e.g., energy / utility transfer and energy / utility balances). They also allow the inclusion of hypothetical modeling configurations based on non-physical relationships.

The UOPSS representation is formed by groups of objects that can be categorized as unit-operation (UO), unit-operation-port-state (UOPS), and unit-operation-port-state-unit-operation-port-state (UOPS-UOPS). The UO groups represent units with operational modes, the UOPS groups additionally include port and state information, and the UOPS-UOPS groups connect adjacent UOPS groups to ensure consistency in the flows throughout the network. The UO groups have binary and continuous variables associated with processing rates, orders, or inventories. The UOPS groups have continuous variables associated with inlet and outlet operations, which are useful for handling multiple inlet or outlet streams within the same unit-operation (e.g., splitters, mixers). The arrows connecting adjacent ports establish the UOPS-UOPS relationships, which are modeled with binary and continuous variables to represent material or energy transfers between unit-operations. An illustrative example providing additional details on the UOPSS representation can be found in Supplementary Material B.

3.2. Indices and sets

The following indices and sets are considered in the refinery scheduling model.

- $i \in \{I, I_{BL}, I_{FT}, I_{TK}, I_{UN}\}$ for inlet ports of unit-operations.
- $j \in \{J, J_{BL}, J_{ST}, J_{TK}, J_{UN}\}$ for outlet ports of unit-operations.

- $m \in M = \{M_{BL}, M_{CDU}, M_{FE}, M_{PR}, M_{TK}, M_{UN}\}$ for unit-operations m of blenders, crude distillation units, feedstocks, products, tanks, and processing units.
- $op \in OP = \{op_1, op_2, \dots, op_{modes}\}$ for operational modes of processing units.
- $p \in \{P_v, P_w\}$ for volume-based and mass-based properties.
- $t, tt \in T = \{t_1, t_2, \dots, t_n\}$ for n time periods.

Fig. 3 presents an illustrative diagram to provide an easier understanding of the relationship among unit-operations and their inter-connected ports. A generic unit-operation m is directly connected to in-port i and out-port j . This inlet port i handles upstream processes to m , i.e., incoming transfers or logistics operations. Moreover, i is connected to j_{up} , which is the out-port of an upstream unit-operation to m , i.e., m_{up} . The arrow connection for the out-port-in-port pair (j_{up}, i) establishes the UOPS-UOPS connection between the unit-operations m_{up} and m . It is similar for the pair (j, i_{do}) that connects unit-operation m with its downstream counterpart m_{do} .

3.3. Subscripts and superscripts

The following subscripts and superscripts are used in the refinery scheduling optimization model.

- BL : blender unit.
- c : crude oil component.
- CDU : crude distillation unit.
- cut : crude oil cutpoint fraction.
- DEV : total flowrate deviation.
- $DEV1$: flowrate deviation (increase).
- $DEV2$: flowrate deviation (decrease).
- do : downstream unit-operation or flow.
- $DRAW$: drawing operations.
- FAC : factor.
- FE : feedstock.
- $FILL$: filling operations.
- FT : feed tank.
- L : lower bound.
- NLP : nonlinear programming.
- op : operational mode of process units.
- opt : optimal value.
- p : property.
- PR : product.
- sg : specific gravity.
- SP : starting point.
- ST : storage tank.
- sul : sulfur concentration.
- TK : tank.
- U : upper bound.
- UN : processing unit.
- up : upstream unit-operation or flow.

3.4. Parameters

The following parameters are included in the refinery scheduling optimization model. They are represented with an overbar to be distinguished from variables. Time-related parameters have values that are multiples of the time-step size.

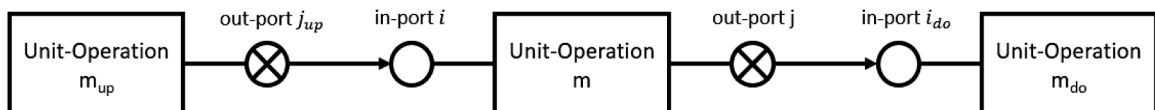


Fig. 3. Relationship among upstream and downstream unit-operations and ports.

- $\overline{\Delta D}_m^L$: lower bound for the fill-draw delay constraint of unit-operation $m \in M_{TK}$.
- $\overline{\Delta D}_m^U$: upper bound for the fill-draw delay constraint of unit-operation $m \in M_{TK}$.
- Δt : time-step size.
- $\overline{dv}^L$: lower bound for a generic decision variable dv .
- $\overline{dv}^{SP}$: starting point of a generic decision variable dv .
- $\overline{dv}^U$: upper bound for a generic decision variable dv .
- $\bar{f}_{i,p,t}$: quality factor for property p from in-port i at time period t .
- $\bar{f}_{j,p,t}$: quality factor for property p to out-port j at time period t .
- $\overline{mu}_{i,t}^L$: lower bound for the multi-use upstream operations to in-port i at time period t .
- $\overline{mu}_{i,t}^U$: upper bound for the multi-use upstream operations to in-port i at time period t .
- $\overline{mu}_{j,t}^L$: lower bound for the multi-use downstream operations from out-port j at time period t .
- $\overline{mu}_{j,t}^U$: upper bound for the multi-use downstream operations from out-port j at time period t .
- $\bar{n}_p$: number of time periods.
- $\overline{pen}_m$: penalty over the performance of distillation unit $m \in M_{CDU}$.
- $\overline{price}_{m,t}$: market value of product $m \in M_{PR}$ or cost of feedstock $m \in M_{FE}$ at time period t .
- $\bar{r}_{i,t}^L$: lower bound for the inverse yield to in-port i at time period t .
- $\bar{r}_{i,t}^U$: upper bound for the inverse yield to in-port i at time period t .
- $\bar{r}_{j,t}^L$: lower bound for the direct yield from out-port j at time period t .
- $\bar{r}_{j,t}^U$: upper bound for the direct yield from out-port j at time period t .
- $\bar{t}_{end}$: time horizon length.
- $\overline{upt}_m^L$: lower bound for the uptime constraint of unit-operation m .
- $\overline{upt}_m^U$: upper bound for the uptime constraint of unit-operation m .
- $\bar{x}_{j,i,t}^L$: lower bound for the material flow from out-port j to in-port i at time period t .
- $\bar{x}_{j,i,t}^U$: upper bound for the material flow from out-port j to in-port i at time period t .
- $\bar{x}_{m,t}^L$: lower bound for the flowrate of unit-operation m at time period t .
- $\bar{x}_{m,t}^U$: upper bound for the flowrate of unit-operation m at time period t .
- $\bar{x}h_{m,t}^L$: lower bound for the inventory of unit-operation $m \in M_{TK}$ at time period t .
- $\bar{x}h_{m,t}^U$: upper bound for the inventory of unit-operation $m \in M_{TK}$ at time period t .
- $\overline{x}h_{m,t}^{DRAW}$: lower bound for the inventory of unit-operation $m \in M_{TK}$ at time period t upon completion of a drawing operation.
- $\overline{x}h_{m,t}^{FILL}$: upper bound for the inventory of unit-operation $m \in M_{TK}$ at time period t upon completion of a filling operation.
- $\overline{x}h_{m,t}^{Max}$: big-M parameter to represent the maximum inventory of unit-operation $m \in M_{TK}$ at time period t .

3.5. Variables

The decision variables considered in the refinery scheduling optimization model comprise quantity (i.e., material streams, inventory levels), logistics (i.e., binary decisions regarding unit-operations and connection flows), and quality (i.e., yields, compositions, properties, conditions) information.

3.5.1. Continuous variables

- dv^{opt} : optimal value of a generic decision variable dv .
- $rand$: random number generated.
- $re_{j,c,cut}^{vie}$: rendering for the yield of crude oil cutpoint cut of crude oil component c in the out-port j .
- $re_{j,c,cut}^p$: rendering for the quality of crude oil cutpoint cut of crude oil component c in the out-port j .
- $v_{i,p,t}$: volume-based quality p of the downstream flow in the in-port i of its respective unit-operation at time period t .
- $v_{j,p,t}$: volume-based quality p of the upstream flow in the out-port j of its respective unit-operation at time period t .
- $w_{i,p,t}$: mass-based property p of the downstream flow in the in-port i of its respective unit-operation at time period t .
- $w_{j,p,t}$: mass-based property p of the upstream flow in the out-port j of its respective unit-operation at time period t .
- $x_{j,i,t}$: material flow from out-port j to in-port i at time period t .
- $x_{j,p,t}^{FAC}$: surplus/slack quality amount of the factor-flow j^{FAC} for each property p at time period t .
- $x_{m,t}$: material flow of unit-operation m at time period t .
- $x_{m,t}^{DEV}$: flowrate deviation in unit-operation $m \in M_{CDU}$ between time periods $t-1$ and t .
- $x_{m,t}^{DEV1}$: flowrate deviation (increase) in unit-operation $m \in M_{CDU}$ between time periods $t-1$ and t .
- $x_{m,t}^{DEV2}$: flowrate deviation (decrease) in unit-operation $m \in M_{CDU}$ between time periods $t-1$ and t .
- $xh_{m,t}$: material inventory of tank unit-operation m at time period t .
- $zsd_{m,t}$: relaxed variable $[0,1]$ to represent shut-down operations of unit-operation m at time period t .
- $zsu_{m,t}$: relaxed variable $[0,1]$ to represent start-up operations of unit-operation m at time period t .
- $zsw_{m,t}$: relaxed variable $[0,1]$ to represent switch-over-to-itself operations of unit-operation m at time period t .
- Z : objective function to be maximized.

3.5.2. Binary variables

- $y_{j,i,t}$: discrete decision for the connection between out-port j and in-port i at time period t .
- $y_{m,t}$: discrete decision associated with unit-operation m at time period t .
- $y_{m^{op},t}$: discrete decision for operational mode op of unit-operation m at time period t .
- $yd_{m,t}^{DRAW}$: discrete decision for drawing operations in unit-operation $m \in M_{TK}$ at time period t .
- $yd_{m,t}^{FILL}$: discrete decision for filling operations in unit-operation $m \in M_{TK}$ at time period t .

3.6. Base formulation

The base formulation is defined in Eqs. (1) to (15) and includes both continuous and binary variables within UO, UOPS, and UOPS-UOPS relationships. All out-ports j and in-ports i are associated with a unit-operation m (i.e., j_m and i_m), but the subscript m is omitted for the sake of simplicity. Eqs. (1a) to (1c) define the non-negative real variables $x_{m,t}$, $xh_{m,t}$, and $x_{j,i,t}$, and Eqs. (2a) to (2b) define the binary variables $y_{m,t}$ and $y_{j,i,t}$.

$$x_{m,t} \geq 0 \quad \forall (m, t) \quad (1a)$$

$$xh_{m,t} \geq 0 \quad \forall (m, t) \quad (1b)$$

$$x_{j,i,t} \geq 0 \quad \forall (j, i, t) \quad (1c)$$

$$y_{m,t} = \{0, 1\} \quad \forall (m, t) \quad (2a)$$

$$y_{j,i,t} = \{0, 1\} \forall (j, i, t) \quad (2b)$$

The arrows throughout the network in Fig. 3 represent the UOPS-UOPS material flows $x_{j,i,t}$, which are constrained by the inlet and outlet flowrate (e.g., hydraulic limits) and by the processing capacities of units. They vary between lower and upper bounds ($\bar{x}_{j,i,t}^L$ and $\bar{x}_{j,i,t}^U$) if their binary variables $y_{j,i,t}$ are active, as shown in Eq. (3). If the binary variable is zero, the flows are automatically set to zero, which indicates that the respective UOPS-UOPS transfer (j, i, t) is not permitted. This could represent maintenance procedures in the downstream or upstream unit connected to the flow $x_{j,i,t}$, or in the pipelines as well. Similarly, Eq. (4) imposes bounds ($\bar{x}_{m,t}^L$ and $\bar{x}_{m,t}^U$) for the UO flows of unit-operations $x_{m,t}$ ($m \in \{M_{UN}, M_{BL}\}$) using binary variables $y_{m,t}$ ($m \in \{M_{UN}, M_{BL}\}$), and Eq. (5) imposes bounds ($\bar{x}h_{m,t}^L$ and $\bar{x}h_{m,t}^U$) for the UO tank holdups or inventory levels $xh_{m,t}$ ($m \in M_{TK}$) using the binary variables $y_{m,t}$ ($m \in M_{TK}$).

$$\bar{x}_{j,i,t}^L y_{j,i,t} \leq x_{j,i,t} \leq \bar{x}_{j,i,t}^U y_{j,i,t} \forall (j, i, t) \quad (3)$$

$$\bar{x}_{m,t}^L y_{m,t} \leq x_{m,t} \leq \bar{x}_{m,t}^U y_{m,t} \forall t, m \in \{M_{BL}, M_{UN}\} \quad (4)$$

$$\bar{x}h_{m,t}^L y_{m,t} \leq xh_{m,t} \leq \bar{x}h_{m,t}^U y_{m,t} \forall m \in M_{TK}, t \quad (5)$$

If the binary variable $y_{m,t}$ of the unit-operation m at time period or step t is active, Eqs. (6) to (9) constrain the summation of the quantity-flows $x_{j,i,t}$ incoming to or outgoing from its port-states. Eqs. (6) and (7) impose lower ($\bar{x}_{m,t}^L$) and upper ($\bar{x}_{m,t}^U$) bounds for the sum of the UOPS-UOPS quantity-flows $x_{j,i,t}$ outgoing from the out-port j_{up} of m_{up} (unit-operation upstream of m) and incoming to the in-ports i of m . Similarly, Eqs. (8) and (9) impose bounds for the sum of the $x_{j,i,t}$ quantity-flows outgoing from the out-ports j of m and incoming to the in-ports i_{do} of m_{do} (unit-operation downstream of m). If $y_{m,t}$ is inactive, the quantity-flows in Eqs. (6) to (9) are set to 0.

$$\sum_{j_{up}} x_{j_{up},i,t} \geq \bar{x}_{m,t}^L y_{m,t} \forall (i, m, t) \quad (6)$$

$$\sum_{j_{up}} x_{j_{up},i,t} \leq \bar{x}_{m,t}^U y_{m,t} \forall (i, m, t) \quad (7)$$

$$\sum_{i_{do}} x_{j,i_{do},t} \geq \bar{x}_{m,t}^L y_{m,t} \forall (m, j, t) \quad (8)$$

$$\sum_{i_{do}} x_{j,i_{do},t} \leq \bar{x}_{m,t}^U y_{m,t} \forall (m, j, t) \quad (9)$$

Unit-operations $m \in \{M_{BL}, M_{UN}\}$ can have multiple streams incoming to or outgoing from their independent connected port-states. Eqs. (10) to (13) introduce bounds on yields, either inverse/input or direct/output. Inverse yields model the mixer-based unit-operations, in which there are multiple inlet streams within a given time window. Direct yields model splitter-based unit-operations, which split the incoming quantities in multiple outlet streams within a given time window. Inverse and direct yields correspond to some structural parameters often used in the resource-task-network (RTN) and state-task-network (STN) representations, and are respectively associated with the consumption intensity of feed resources and production intensity of product resources in unit-operations (Castro et al., 2018). This set of constraints is useful to configure yields for inlet and outlet flows. For flows $x_{m,t}$ of unit-operation $m \in M_{BL}$, Eqs. (10) and (11) define bounds of inverse yields ($\bar{r}_{i,t}^L$ and $\bar{r}_{i,t}^U$) for the upstream operations to m (e.g., to constrain the percentage yields of streams to be blended). For throughputs $x_{m,t}$ of unit-operation $m \in M_{UN}$, Eqs. (12) and (13) define bounds of direct yields ($\bar{r}_{j,t}^L$ and $\bar{r}_{j,t}^U$) for downstream operations to m (e.g., specifying yields of unit outputs). Eqs. (12) and (13) are applied for all unit-operations, given that they behave similarly to splitter units, i.e., with a single input stream but potentially multiple output streams. The yields $\bar{r}_{i,t}^L$ and

$\bar{r}_{i,t}^U$ can be configured to be fixed (i.e., $\bar{r}_{i,t}^L = \bar{r}_{i,t}^U$) or to provide an optimizable range (i.e., variable recipe) as an additional degree of freedom (i.e., $\bar{r}_{i,t}^L < \bar{r}_{i,t}^U$). Depending on the process or production network configuration, Eqs. (10) and (11) can be extended to include splitter-based units, and Eqs. (12) and (13) can be extended to include mixer-based units. It is worth noting that Eqs. (10) to (13) are not valid for tanks, which typically have only one active inlet flow and only one active outlet flow at the same time window (i.e., due to operational limitations, tanks cannot have multiple inlet flows or multiple outlet flows simultaneously).

$$\sum_{j_{up}} x_{j_{up},i,t} \geq \bar{r}_{i,t}^L x_{m,t} \forall (i, t), m \in \{M_{BL}\} \quad (10)$$

$$\sum_{j_{up}} x_{j_{up},i,t} \leq \bar{r}_{i,t}^U x_{m,t} \forall (i, t), m \in \{M_{BL}\} \quad (11)$$

$$\sum_{i_{do}} x_{j,i_{do},t} \geq \bar{r}_{j,t}^L x_{m,t} \forall (j, t), m \in \{M_{UN}\} \quad (12)$$

$$\sum_{i_{do}} x_{j,i_{do},t} \leq \bar{r}_{j,t}^U x_{m,t} \forall (j, t), m \in \{M_{UN}\} \quad (13)$$

Eq. (14) represents the material balance required to calculate the inventory or holdup $xh_{m,t}$ of storage and feed tanks $m \in M_{TK}$. At each time window t , the holdup amount $xh_{m,t}$ is the remaining amount of material in the past time period ($xh_{m,t-1}$), increased by the incoming material to the tank from the upstream connections ($x_{j_{up},i,t}$) and subtracted by the outgoing material from the downstream connections from the tank ($x_{j,i_{do},t}$).

$$xh_{m,t} = xh_{m,t-1} + \sum_{j_{up}} x_{j_{up},i,t} - \sum_{i_{do}} x_{j,i_{do},t} \quad (14)$$

$$\forall (i, j, t), m \in M_{TK}$$

The material balances to impose no accumulation of material in continuous processes are defined in Eq. (15) for the unit-operation $m \in \{M_{BL}, M_{UN}\}$, whereby the total incoming flow to the unit-operation equals its total outlet flow in the same time period or step. The pairs (j_{up}, i) and (j, i_{do}) are the upstream and downstream connections of m .

$$\sum_{j_{up}} x_{j_{up},i,t} = \sum_{i_{do}} x_{j,i_{do},t} \quad (15)$$

$$\forall ((j_{up}, i), (j, i_{do}), t), m \in \{M_{BL}, M_{UN}\}$$

3.7. Logistics mathematical formulation: MILP refinery scheduling

Eqs. (16) and (17) are structural transition constraints that coordinate setups of connected unit-operations through the out-port j_{up} of m_{up} and the in-port i of m . If the binary variables of a given pair of unit-operations (m_{up}, m) are active, then the binary variables $y_{j,i,t}$ that represents their connection may be active (i.e., material streams are allowed between them). Thus, active unit-operations imply active port-states connecting them. However, if at least one of the units is not operating (i.e., binary variable equal to zero), there cannot be a material stream connecting these unit-operations (i.e., ports are not active). This logic valid cut is helpful to reduce search in branch-and-bound algorithms and is the disaggregated form where the sum of Eqs. (16) and (17) is the aggregated form and removes a constraint but is less tight.

$$y_{m_{up},t} \geq y_{j_{up},i,t} \forall (m_{up}, j_{up}, i, t) \quad (16)$$

$$y_{m,t} \geq y_{j_{up},i,t} \forall (m, j_{up}, i, t) \quad (17)$$

Eq. (18) ensures that at most one operational mode op is simultaneously allowed for each physical unit m at each time period t . Eq. (19) correlates each unit-operation m with its respective operational modes op , so that if the unit-operation is active, one operational mode must be selected at each time window. Eq. (20) is the zero-downtime constraint

to ensure the continuous selection of a mode of operation op , which is useful for unit-operations that must be continuously operated for all periods within the time horizon, such as CDUs.

$$\sum_{op \in OP} y_{m^{op},t} \leq 1 \quad \forall t, m \in \{M_{UN}\} \quad (18)$$

$$y_{m,t} = \sum_{op \in OP} y_{m^{op},t} \quad \forall t, m \in \{M_{UN}\} \quad (19)$$

$$\sum_{op \in OP} y_{m^{op},t} = 1 \quad \forall t, m \in \{M_{CDU}\} \quad (20)$$

The temporal transition constraints represented by Eqs. (21) to (23) coordinate the operation of the **semi-continuous blender units** (Kelly and Zyngier, 2007; Zyngier and Kelly, 2009). The binary variable $y_{m,t}$ manages the variables related to the start-up ($zsu_{m,t}$), shut-down ($zsd_{m,t}$), and switch-over-to-itself ($zsw_{m,t}$) operations, which are respectively associated with starting, shutting down, or changing the operational mode of a blender unit-operation. In the proposed formulation, the temporal transition constraints are considered only for blender unit-operations because the processing unit-operations $m \in M_{UN}$ are assumed to continuously operate. However, such constraints can be useful for handling non-continuous or semi-continuous operations and to introduce scheduled maintenance for the physical units.

$$y_{m,t} - y_{m,t-1} - zsu_{m,t} + zsd_{m,t} = 0 \quad \forall m \in M_{BL}, t \quad (21)$$

$$y_{m,t} + y_{m,t-1} - zsu_{m,t} - zsd_{m,t} - 2zsw_{m,t} = 0 \quad \forall m \in M_{BL}, t \quad (22)$$

$$zsu_{m,t} + zsd_{m,t} + zsw_{m,t} \leq 1 \quad \forall m \in M_{BL}, t \quad (23)$$

Eqs. (24) and (25) are multi-use procedure constraints, in which downstream in-ports i_{do} connected to the out-ports j are limited by lower and upper bounds $\overline{mu}_{j,t}^L$ and $\overline{mu}_{j,t}^U$. This is helpful to control the number of simultaneous drawing operations allowed from a blender $m \in M_{BL}$ to its downstream tanks. Similarly, (26) and (27) impose bounds $\overline{mu}_{i,t}^L$ and $\overline{mu}_{i,t}^U$ for the upstream out-ports j_{up} connected to the in-ports i , which is helpful to control the number of simultaneous filling operations to unit-operations $m \in M_{UN}$.

$$\sum_{i_{do}} y_{j,i_{do},t} \geq \overline{mu}_{j,t}^L y_{m,t} \quad \forall (j, m \in M_{BL}, t) \quad (24)$$

$$\sum_{i_{do}} y_{j,i_{do},t} \leq \overline{mu}_{j,t}^U y_{m,t} \quad \forall (j, m \in M_{BL}, t) \quad (25)$$

$$\sum_{j_{up}} y_{j_{up},i,t} \geq \overline{mu}_{i,t}^L y_{m,t} \quad \forall (i, m \in M_{UN}, t) \quad (26)$$

$$\sum_{j_{up}} y_{j_{up},i,t} \leq \overline{mu}_{i,t}^U y_{m,t} \quad \forall (i, m \in M_{UN}, t) \quad (27)$$

Eqs. (28) and (29) model the **uptime or run-length constraints** from Kelly and Zyngier (2007) in which $\overline{upt}_m^L$ and $\overline{upt}_m^U$ are the respective uptime lower and upper bounds, tt is an auxiliary time index, $\bar{t}_{end}$ is the time horizon length, and Δt is the time step. They impose operational time limits for unit-operations m , so that minimum and maximum amounts of time can be configured (i.e., upon starting an operation, a given unit cannot operate less than the uptime lower bound or more than the uptime upper bound). In Eq. (28), if a unit is operating at time window t , there cannot have been more than one start-up operation within a past time range $\overline{upt}_m^L$; and if the unit is not operating, no start-up operation within a past time range $\overline{upt}_m^L$ is allowed. Eq. (29) imposes that unit-operations m cannot continuously operate over a time range higher than the upper uptime bound of $\overline{upt}_m^U$. Eq. (30) models a unit-operation uptime temporal aggregation cut. Given the uptime lower bound,

number of time periods $\bar{n}_p$, and time step size, there is a limit in the number of possible start-up operations that can be performed over the time horizon length. Eq. (30) reduces the search space of the start-up variable $zsu_{m,t}$, whereby the global optimum region is guaranteed to remain feasible after the cut. It is worth noting that Eqs. (28) to (30), combined with Eqs. (21) to (23), can properly manage the correlation between the variables $zsu_{m,t}$, $zsd_{m,t}$, and $zsw_{m,t}$ and the operational transitions for unit-operations m .

$$\left(\frac{\overline{upt}_m^L}{\Delta t} - 1 \right) \sum_{tt=1 | tt < t} zsu_{m,t-tt} \leq y_{m,t} \quad \forall (m, t \geq 2) \quad (28)$$

$$\sum_{tt=t}^{\left(t + \frac{\overline{upt}_m^U}{\Delta t} \right)} y_{m,tt} \leq \frac{\overline{upt}_m^U}{\Delta t} \quad \forall \left(m, t \leq \bar{t}_{end} - \frac{\overline{upt}_m^U}{\Delta t} \right) \quad (29)$$

$$\sum_t zsu_{m,t} \leq \left(\frac{\bar{n}_p \Delta t}{\overline{upt}_m^L + \Delta t} \right) \quad \forall m \quad (30)$$

Eqs. (31) and (32) manage the minimum ($\overline{\Delta D}_m^L$) and maximum ($\overline{\Delta D}_m^U$) time between the latest filling and the next drawing operations, referred to as the **fill-draw delay**, for the upstream j_{up} and downstream i_{do} connections of a tank $m \in M_{TK}$. In Eq. (31), whenever there is an incoming flow to a tank, any outgoing flow from that tank is only allowed after a waiting period of $\overline{\Delta D}_m^L$. Additionally, Eq. (31) ensures that filling and drawing operations cannot be simultaneously performed to the unit-operation $m \in M_{TK}$ at time window t , which represents a logistics constraint commonly used in the plant. In Eq. (32), whenever there is an incoming flow to a tank, there must be at least one outgoing flow from that tank within the period $\overline{\Delta D}_m^U$. These constraints are useful for handling materials that require a **decantation period** prior to their processing; and for **avoiding long storage periods** of materials.

$$y_{m,j_{up},i,t} + y_{m,j,i_{do},t+tt} \leq 1 \quad \forall (j_{up}, i, j, i_{do}), m \in M_{TK}, tt = 0.. \overline{\Delta D}_m^L, t = 1.. \bar{t}_{end}, t + tt \leq \bar{t}_{end} \quad (31)$$

$$y_{m,j_{up},i,t} - \sum_{tt=1}^{\overline{\Delta D}_m^U} y_{m,j,i_{do},t+tt} \leq 0 \quad \forall (j_{up}, i, j, i_{do}), m \in M_{TK}, t + tt \leq \bar{t}_{end} \quad (32)$$

Eqs. (33) to (36) represent the fill-to-full and draw-to-empty operations for tanks. Eq. (33) introduces the logic variable $yd_{m,t}^{FILL}$, which is active when there is a filling operation. Similarly, Eq. (34) introduces the logic variable $yd_{m,t}^{DRAW}$ for drawing operations. Eq. (35) enforces that once a filling operation starts, there is a minimum inventory $\overline{xh}_{m,t}^{FILL}$ that must be reached before allowing any drawing operation from the same tank. Similarly, whenever there is a drawing operation, Eq. (36) imposes a maximum inventory $\overline{xh}_{m,t}^{DRAW}$ that must be reached before any filling operation is allowed to the same tank. The coefficients $\overline{xh}_{m,t}^{FILL}$ and $\overline{xh}_{m,t}^{DRAW}$ are the fill-to-full and the draw-to-empty inventories, which may be static or dynamic over the time horizon. Eqs. (35) and (36) use the maximum inventory available $\overline{xh}_{m,t}^{Max}$ at each time window t to control the auxiliary binary variables using big-M constraints (see Winston and Goldberg, 2004 for detailed information on big-M methods). Thus, filling operations continuously fill the tank at least until it reaches a minimum inventory $\overline{xh}_{m,t}^{FILL}$, and drawing operations must be carried out at least until the tank reaches a maximum inventory $\overline{xh}_{m,t}^{DRAW}$. This is especially helpful for reducing the number of operations involving tanks (i.e., for improved industrial operations, it is beneficial to perform fewer

operations in larger amount rather than multiple operations in smaller amount). Such constraints are alternatives to the commonly employed changeover costs, which are arbitrary limitations to account for implicit costs of processing operations (e.g., opening valves).

$$y_{m,j_{up},i,t} \leq y_{m,t}^{FILL} \quad \forall (j_{up}, i), m \in M_{TK}, t \quad (33)$$

$$y_{m,j_{do},t} \leq y_{m,t}^{DRAW} \quad \forall (j, i_{do}), m \in M_{TK}, t \quad (34)$$

$$\left(\overline{xh}_{m,t-1}^{FILL} - xh_{m,t-1} \right) \leq y_{m,t}^{FILL} \overline{xh}_{m,t}^{Max} \quad \forall m \in M_{TK}, t \geq 2 \quad (35)$$

$$\left(\overline{xh}_{m,t-1}^{DRAW} - xh_{m,t-1} \right) \geq -y_{m,t}^{DRAW} \overline{xh}_{m,t}^{Max} \quad \forall m \in M_{TK}, t \geq 2 \quad (36)$$

To improve the accuracy of the MILP formulation, we utilize a linear approximation for nonconvex blending constraints proposed by Kelly et al. (2018) and shown in Eq. (37). For each property p in the blender $m \in M_{BL}$, amounts of feedstocks $\sum_{j_{up}} x_{j_{up},i,t}$ incoming from in-ports i with

associated quality factors $\bar{f}_{i,p,t}$ counterbalance the overall quality $\sum_{i_{do}} x_{j,i_{do},t}$

of blended material with quality factors $\bar{f}_{j,p,t}$. The slack or surplus variable $x_{j^{FAC},p,t}$ represents the insufficient or exceeded quality amount for property p . For an upper bound of property specification, $x_{j^{FAC},p,t} \leq 0$ and for a lower bound, $x_{j^{FAC},p,t} \geq 0$.

$$\sum_{i \in I_{BL}} \bar{f}_{i,p,t} \sum_{j_{up} \in J_{ST}} x_{j_{up},i,t} = \bar{f}_{j,p,t} \sum_{i_{do} \in I_{TT}} x_{j,i_{do},t} + x_{j^{FAC},p,t} \quad \forall j, j^{FAC}, p, t \quad (37)$$

3.8. Quality mathematical formulation: NLP refinery scheduling

For the quality balances throughout the network, we consider p as property (component concentration, specific gravity, sulfur content) in which v and w are the volume- and mass-based properties, respectively as described by Kelly and Zyngier (2017). The variables $v_{j_{up},p,t}$ and $w_{j_{up},p,t}$ are the volume- and mass-based qualities of the upstream flow from the out-ports j_{up} (of unit-operation m_{up}) incoming to the in-port i (of unit-operation m), whereby $v_{j_{up},p=sg,t}$ refers specifically for specific gravity; and $v_{i,p,t}$ and $w_{i,p,t}$ are the volume- and mass-based qualities of inlet flows $i \in I$. It is similar for the downstream variables $v_{i_{do},p,t}$ and $w_{i_{do},p,t}$. Fig. 4 illustrates the quality variables associated with the inlet and outlet ports. It is worth mentioning that the quality at a given port-state is also the quality of the material flow outgoing from this port-state (i.e., assumption of perfect mixing).

Eq. (38) is an overall mass-balance at the in-port $i \in \{I_{BL}, I_{UN}\}$ of mixer-based and splitter-based unit-operations $m \in \{M_{BL}, M_{UN}\}$. It introduces the volume-based properties $p \in P_v$ such as components (crude oil compositions) and densities (specific gravity). Similarly, Eq. (39) is a property mass-balance (e.g., sulfur concentration), which introduces the mass-based properties $p' \in P_w$ and is trilinear, versus bilinear for the volume-based property, due to the inclusion of the known and assigned density or specific-gravity volume-based property variable.

$$v_{i,p,t} \sum_{j_{up}} x_{j_{up},i,t} = \sum_{j_{up}} v_{j_{up},p,t} x_{j_{up},i,t} \quad \forall t, i \in \{I_{BL}, I_{UN}\}, p \in P_v \quad (38)$$

$$w_{i,p',t} \sum_{j_{up}} v_{j_{up},p,t} x_{j_{up},i,t} = \sum_{j_{up}} w_{j_{up},p',t} v_{j_{up},p,t} x_{j_{up},i,t} \quad \forall t, i \in \{I_{BL}, I_{UN}\}, p \in P_v, p' \in P_w \quad (39)$$

Eqs. (40) and (41) represent the quality balances for volume- and mass-based properties in tanks. They are a combination of Eq. (14), that ensures consistency for the material-balance in tanks, with Eqs. (38) and (39) for the overall and component mass-balances when mixing two or more streams. The subsets in the summations involving the in-ports i and out-ports j in Eqs. (40) and (41) are also omitted, since this is valid for all $(i, m, j) \in M_{TK}$, by which j_{up} represents the upstream out-ports of the connected unit-operations m_{up} arriving in the in-port $i \in I_{TK}$, and i_{do} represents the downstream in-ports of the connected unit-operations m_{do} outgoing from the out-port $j \in J_{TK}$. The quality variable for the out-ports of a tank unit-operation ($m \in M_{TK}$) is the quality of the blend within the tank, as defined by Eqs. (42) and (43) for volume- and mass-based properties, respectively. These two equations represent the implicit or inherent splitter constraints when the tanks are modeled assuming perfectly-mixed accumulation.

$$v_{m,p,t} xh_{m,t} = v_{m,p,t-1} xh_{m,t-1} + \sum_{j_{up}} v_{j_{up},p,t} x_{j_{up},i,t} - v_{m,p,t} \sum_{i_{do}} x_{j,i_{do},t} \quad (40)$$

$$\forall t, (i, m, j) \in \{I_{TK}, M_{TK}, J_{TK}\}, p \in P_v$$

$$w_{m,p',t} v_{m,p,t} xh_{m,t} = w_{m,p',t-1} v_{m,p,t-1} xh_{m,t-1} + \sum_{j_{up}} w_{j_{up},p',t} v_{j_{up},p,t} x_{j_{up},i,t} - w_{m,p',t} v_{m,p,t} \sum_{i_{do}} x_{j,i_{do},t} \quad (41)$$

$$\forall t, (i, m, j) \in \{I_{TK}, M_{TK}, J_{TK}\}, p \in P_v, p' \in P_w$$

$$v_{j,p,t} = v_{m,p,t} \quad \forall (m, j) \in \{M_{TK}, J_{TK}\}, p \in P_v, t \quad (42)$$

$$w_{j,p',t} = w_{m,p',t} \quad \forall (m, j) \in \{M_{TK}, J_{TK}\}, p' \in P_w, t \quad (43)$$

Eqs. (44), (45), and (46) respectively are the overall material-balance, overall mass-balance, and component mass-balance for the outputs of unit-operations based on distillation processes. Eq. (44)

converts the CDU throughputs $\left(\sum_{j_{up}} x_{j_{up},i_{CDU},t} \right)$ from the i_{CDU} inlet into

distillate yields $\left(\sum_{i_{do}} x_{j,i_{do},t} \right)$ outgoing from the CDU out-ports j to the

connected downstream in-ports i_{do} . This is performed for all crude oil components c (from the crude oil assay) over all pre-defined distillation fractions cut by introducing the rendering or recovering variable $re_{j,c,cut}^{ye}$, which provides the yield value for each set of (j, c, cut) . Similarly, Eqs. (45) and (46) respectively calculate the volume- and mass-based properties. The rendering variable for qualities, $re_{j,c,cut}^p$, can be accordingly adapted for volume-based qualities (e.g., $re_{j,c,cut}^{p=sg}$, in which the superscript sg stands for specific gravity) and mass-based qualities (e.g., $re_{j,c,cut}^{p=sul}$, in which the superscript sul stands for sulfur concentration).

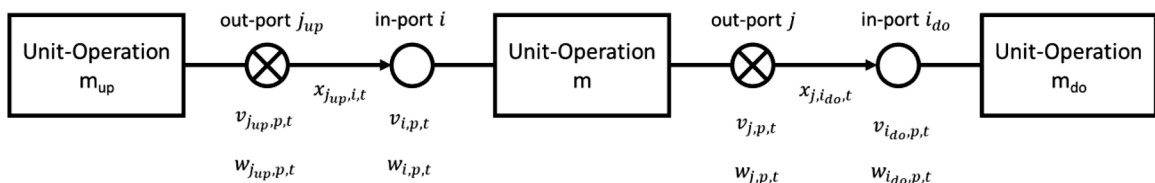


Fig. 4. Quality variables for unit-operations and port-states within the UOPSS representation.

$$\sum_{i_{do}} x_{j,i_{do},t} = \sum_{j_{up}} x_{j_{up},i_{CDU},t} \sum_c \sum_{cut} v_{i_{CDU},p=c,t} r_{j,c,cut}^{vie} \quad \forall (j, t) \quad (44)$$

$$v_{j,p,t} \sum_{i_{do}} x_{j,i_{do},t} = \sum_{j_{up}} x_{j_{up},i_{CDU},t} \sum_c \sum_{cut} v_{i_{CDU},p=c,t} r_{j,c,cut}^{vie} r_{j,c,cut}^p \quad \forall (j, t), p \in P_v \quad (45)$$

$$\begin{aligned} & v_{j,p,t} w_{j,p',t} \sum_{i_{do}} x_{j,i_{do},t} \\ &= \sum_{j_{up}} x_{j_{up},i_{CDU},t} \sum_c \sum_{cut} v_{i_{CDU},p=c,t} r_{j,c,cut}^{vie} r_{j,c,cut}^p r_{j,c,cut}^{p'} \quad (46) \\ & \forall (j, t), p \in P_v, p' \in P_w \end{aligned}$$

For material- and mass-balances of other unit-operations $m \in M_{UN}$, the mathematical equations depend on the model employed for calculating the input-output correlation. Such models can be based on fixed non-composition-based yields (e.g., regardless of the input flow composition, there is a fixed set of yields used for calculating each one of the pre-defined outputs), fixed composition-based yields (i.e., similar to the distillation model in Eqs. (44) to (46), in which the outlet flows of distillates depend on the composition of the crude oil blend incoming to the CDU), process-based yields (i.e., that depend on a set of processing variables such as temperature and pressure) which can be composition-based or not, or any other representation that can accurately estimate the unit-operation outputs. In this work, the remaining processing units consider simplified correlations based on **non-composition yields** or **process-based yields** that depend on processing variables such as temperature, pressure, etc. known as operating conditions.

3.9. Objective function for the refinery scheduling optimization

The objective function for the refinery scheduling optimization in Eq. (47) **maximizes the product revenues** by subtracting the feedstock costs and a performance term for the CDU throughputs to minimize process fluctuations.

$$\text{Max } Z = \sum_t \left(\sum_{m \in M_{PR}} \overline{\text{price}}_{m,t} x_{m,t} - \sum_{m \in M_{FE}} \overline{\text{price}}_{m,t} x_{m,t} - \sum_{m \in M_{CDU}} \overline{\text{pen}}_m x_{m,t}^{DEV} \right) \quad (47)$$

Variable Z is the total objective function to be maximized over the entire time horizon with time periods t , $\overline{\text{price}}_{m,t}$ is the market value or estimated intermediate stream worth of each product $m \in M_{PR}$ or the cost of each feedstock $m \in M_{FE}$, $x_{m,t}$ are the amounts or flows of the respective products or feedstocks, $\overline{\text{pen}}_m$ is a penalty parameter introduced to manage the performance (i.e., for improved operating conditions) of distillation units $m \in M_{CDU}$, and $x_{m,t}^{DEV}$ is the **deviation between the amount of crude processed in the distillation unit over consecutive time periods**. In industrial operations, the operational conditions are expected to be as smooth as possible, without abrupt changes or variations that may compromise the process control within such highly dynamic environments. This performance term smooths the CDU throughputs $x_{m,t}$ ($m \in M_{CDU}$) by calculating the variations of its adjacent amounts and minimizing the linear deviation of the flow in consecutive time periods. Then, $x_{m,t}^{DEV} = |x_{m,t} - x_{m,t-1}|$. We propose Eqs. (48) to (52) as a linear reformulation of such constraint, which introduce the terms $x_{m,t}^{DEV1}$ and $x_{m,t}^{DEV2}$ to represent penalties when the CDU flowrate changes, and constrain them accordingly. This is valid because $x_{m,t}^{DEV}$ is minimized in the objective function. The upper bound $\bar{x}_{m,t}^U$ is the maximum flowrate of unit-operation $m \in M_{CDU}$ at time period t .

$$x_{m,t}^{DEV} = x_{m,t}^{DEV1} + x_{m,t}^{DEV2} \quad \forall t \geq 2, m \in M_{CDU} \quad (48)$$

$$x_{m,t}^{DEV1} \geq x_{m,t} - x_{m,t-1} \quad \forall t \geq 2, m \in M_{CDU} \quad (49)$$

$$x_{m,t}^{DEV2} \geq x_{m,t-1} - x_{m,t} \quad \forall t \geq 2, m \in M_{CDU} \quad (50)$$

$$0 \leq x_{m,t}^{DEV1} \leq \bar{x}_{m,t}^U \quad \forall t \geq 2, m \in M_{CDU} \quad (51)$$

$$0 \leq x_{m,t}^{DEV2} \leq \bar{x}_{m,t}^U \quad \forall t \geq 2, m \in M_{CDU} \quad (52)$$

4. Optimization of the industrial-scale refinery scheduling problem

The second main contribution of this work is the development of a state-of-the-art decision-making framework based on modeling and optimization algorithms to effectively solve industrial-scale refinery scheduling. Fig. 5 provides a conceptual diagram that illustrates the core concepts employed in the proposed framework, which is introduced in Fig. 6. In addition to the novel mathematical formulation developed for the refinery scheduling optimization problem, the framework comprises a combination of computer-aided approaches such as phenomenological (quantity, logic and quality) and temporal decompositions, linear reformulation of nonconvex blending equations, surrogate modeling techniques to represent complex processing units, a novel heuristic algorithm based on relaxation methods, a novel solving method to tackle NLP models, and rescheduling mechanisms.

In the following we present a pseudo-code of the algorithm with further information on the capabilities employed within the framework.

1) Call Pre-processing routine

Define parameters and hyper-parameters required (e.g., scheduling parameters, optimization gap for commercial solvers, tolerances, convergence criteria, etc.).
Read incumbent data to establish initial process conditions (e.g., actual inventories, qualities, etc.).

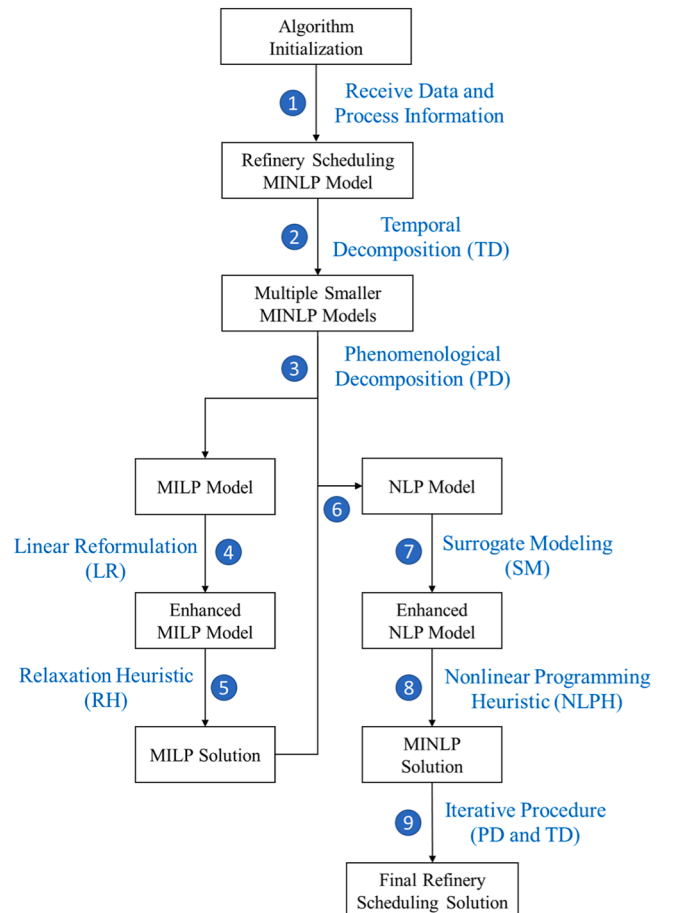


Fig. 5. Conceptual diagram for the proposed framework.

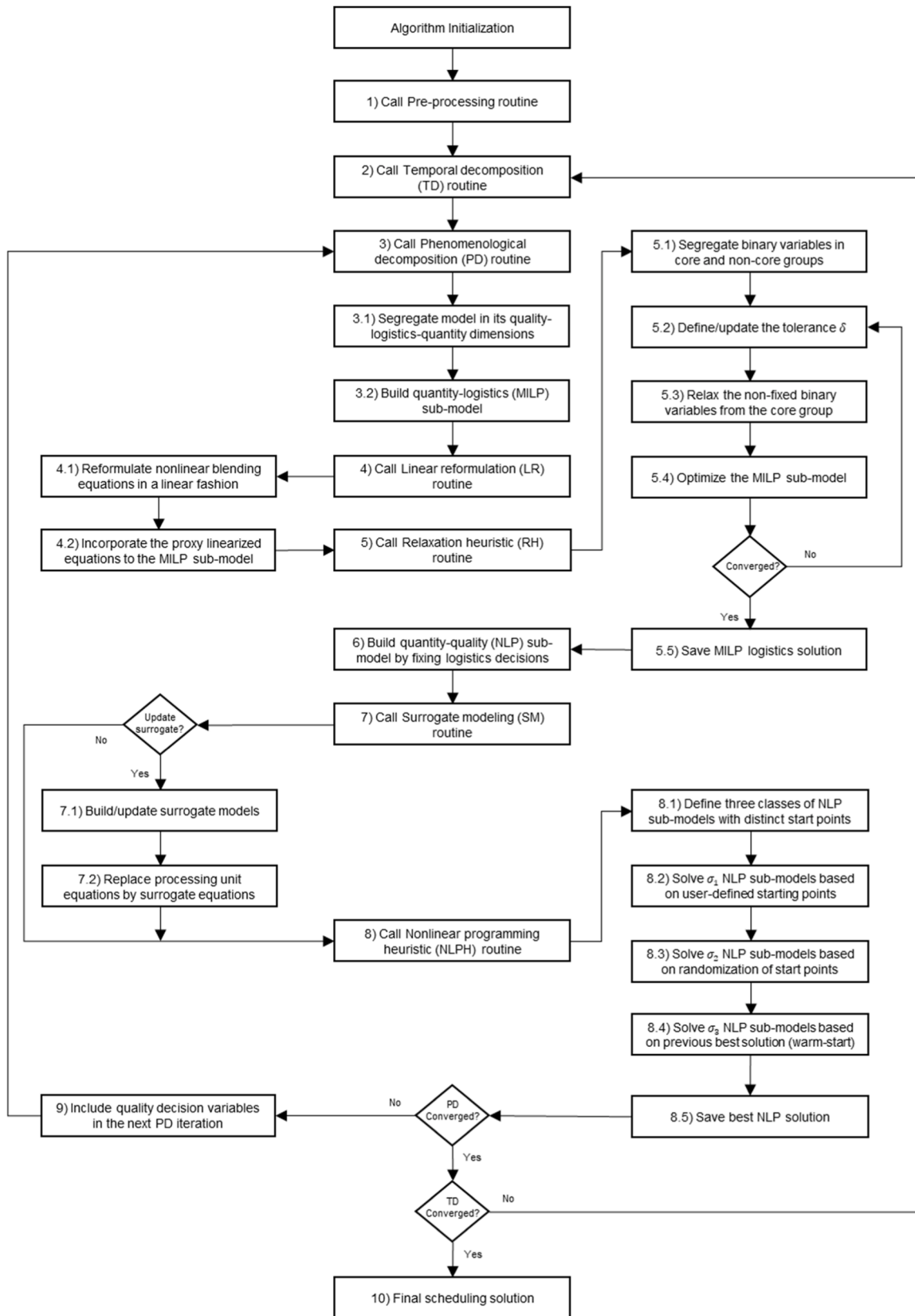


Fig. 6. Modeling and optimization framework for industrial-scale refinery scheduling.

- 2) **Call Temporal Decomposition (TD) routine**
 - 2.1) Segregate the time horizon into multiple time chunks to be sequentially solved. The initial process conditions (i.e., inventories, qualities) of time chunk $tc + 1$ are the respective process conditions at the end of time chunk tc .
- 3) **Call Phenomenological Decomposition (PD) routine**
 - 3.1) Segregate the original MINLP model according to its quantity-logistics-quality phenomena into MILP (quantity and logistics) and NLP (quantity and quality) sub-models.
 - 3.2) Build quantity-logistics (MILP) sub-model by fixing/neglecting quality information, which ensures the variables, constraints, and objective function to be necessarily linear.
- 4) **Call Linear Reformulation (LR) routine**
 - 4.1) Reformulate nonlinear blending equations in a linear fashion by introducing hypothetical slack/surplus variables. Such a heuristic procedure is applied only for the crude oil blending equations.
 - 4.2) Incorporate the linearized equations into the MILP sub-model to provide more accurate predictions.
- 5) **Call Relaxation Heuristic (RH) routine**
 - 5.1) Segregate binary variables into core and non-core groups.
 - 5.2) Define or update the tolerance δ .
 - 5.3) Relax the non-fixed binary variables from core group.
 - 5.4) Optimize the MILP sub-model. If convergence criteria are met, go to 5.5. Otherwise, go back to 5.2.
 - 5.5) Save incumbent MILP logistics solution (binary variables).
- 6) **Build NLP sub-model**
 - 6.1) Build the quantity-quality NLP sub-model by fixing the previously saved logistics solution in the MINLP model.
- 7) **Call Surrogate Modeling (SM) routine**
 - 7.1) Build or update surrogates according to a pre-defined criterion (upon changes in the distillation feed recipe).
 - 7.2) Replace processing unit equations by surrogate equations in the NLP sub-model.
- 8) **Call Nonlinear Programming Heuristic (NLPH) routine**
 - 8.1) Define three classes of NLP sub-models.
 - 8.2) Solve σ_1 NLP sub-model based on user-defined starting points (e.g., personal knowledge or experience).
 - 8.3) Solve σ_2 NLP sub-models based on randomization of starting points.
 - 8.4) Solve σ_3 NLP sub-models using a warm-start procedure (the starting points are randomly generated around the optimal values of the best NLP solution found so far).
 - 8.5) The NLP solutions are sorted according to a performance factor (chosen as the objective function value) and the solution with best performance is saved. While the PD routine does not converge, the algorithm goes to Step 9 and subsequently returns to Step 3. While the TD routine does not converge, the algorithm returns to Step 2.
- 9) **Include quality information in the next PD iteration**

While the phenomenological decomposition does not converge, quality decision variables from the incumbent NLP solution are saved and updated into the MILP problem of the next PD iteration in Step 3.
- 10) **Final Scheduling Solution**

Upon convergence of the temporal decomposition, the algorithm provides a solution for the original MINLP problem. Within each PD iteration, the best solution is selected and saved. Each TD iteration originates a piece of a solution, whereas the final scheduling solution is comprised of one or multiple pieces depending on the time horizon segregation in the TD routine.

Some considerations with respect to the methods used in the framework are further discussed in the following. Prior to the algorithm initialization, we highlight the importance of coherent process topology

as well as the selection of additional features such as model representation, scheduling parameters, and tuning hyper-parameters that are relevant to achieve more efficient formulations.

UOPSS Representation: It has been used in many works (Kelly, 2005; Zyngier and Kelly, 2012; Kelly et al., 2017; Brunaud et al., 2020; Zyngier et al., 2018) for large-scale applications and provides efficient capabilities for handling large number of variables and constraints. In fact, despite the introduction of auxiliary variables and constraints (which are not mandatory), it is apparent in many problems that the UOPSS leads to more tractable formulations that can be more easily and quickly solved by optimization solvers.

Scheduling parameters: An important consideration concerns the parameters to be determined prior to solving the scheduling optimization problem, including the time horizon length, time-step or period size, number of time periods. Extensive testing was performed in this work with distinct scheduling parameters to evaluate the interplay among model size, optimization tractability, and solution quality.

Rescheduling: The scheduling parameters play a key role in providing easier and smoother integration of the mathematical optimization core with the schedule implementation in the plant. The proposed framework enables a rescheduling mechanism that exploits the re-optimization in an online moving horizon fashion, in which plant data define the initial process conditions for the incumbent optimization. The idea of rescheduling has been previously discussed in Franzoi et al. (2021a) and provides significant operational benefits and economic gains.

Algorithm hyper-parameters: There are hyper-parameters to be tuned in order to achieve efficient optimization of such large-scale and complex formulations. They are mostly associated with the modeling and solving strategies employed within the framework. This includes the convergence criteria for the phenomenological decomposition, time discretization in the temporal decomposition, segregation of variables and choice of tolerance in the relaxation heuristic, surrogate updating criteria in the surrogate modeling routine, MILP optimization gap, and the number of sequential NLP optimizations. In the following we present a list of hyper-parameters with further considerations and proper explanation for their respective choices.

- $\rho^{PD} = 3$: First convergence criterion (maximum number of iterations) in the phenomenological decomposition algorithm.
- $\phi^{PD} = 0.1\%$: Second convergence criterion (minimum solution improvement from iteration t to iteration $t + 1$) in the phenomenological decomposition algorithm.
- $\tau^{TD} = [1, 10]$: Number of time chunks from the temporal decomposition. Distinct values of τ^{TD} are considered over multiple scenarios.
- $\delta^{NLP} = \{0.025, 0.05, 0.10\}$: Range used for the starting points of decision variables in the third class of NLP optimizations.
- $\delta^{RH} = \{0.01, 0.05, 0.10\}$: Tolerance in the relaxation heuristic. It starts assuming an initial value of 0.01, and reaches 0.10 over three iterations.
- $\sigma^{SM} = \{0, 1\}$: Binary variable to represent the need for surrogate model updating. This is triggered whenever there is arrival of new feedstocks, change in the qualities of feedstocks, or changes in the process conditions that impact the respective processing unit predictions.
- $\epsilon^{MILP} = 1.0\%$: Optimization gap in the MILP optimizations.
- $\eta^{NLP} = \{1, 3, 3\}$: Number of sequential optimizations in each class of NLP sub-models.

Temporal decomposition: The time horizon is uniformly divided into an arbitrary number τ^{TD} of time chunks defined by the user (Kelly, 2002). The model is accordingly divided into τ^{TD} sub-models, which are sequentially optimized. There is no overlap in the horizon of adjacent time chunks, so that the optimal solution of the model from time chunk tc (e.g., final inventories) is used as the start values (e.g., opening inventories) of the model from time chunk $tc + 1$. Extensive testing was

performed to evaluate the interplay among the model reduction from the temporal decomposition, the gains in terms of computational tractability, and the loss in terms of solution quality.

Phenomenological decomposition: This is similar to the heuristic developed in Menezes et al. (2015), which is shown to be appropriate for crude oil refinery scheduling problems. The MINLP model is partitioned and sequentially solved within an iterative procedure, which provides a solution approach for such an intractable large-scale nonconvex scheduling problem.

Linear reformulation: This linear reformulation of nonconvex blending equations has been shown in previous work (Kelly et al., 2018) to significantly improve the solution quality of the MILP sub-models and to reduce the MILP-NLP decomposition gap. It provides updated quality information to the MILP sub-model, which leads to more quality realistic logistics solutions as quality infeasible portions of the search space can be cutoff in the logistics sub-solves.

Relaxation heuristic: We propose a relaxation heuristic (RH) to tackle large-scale MIP problems. The number of binary variables is significantly reduced whereas sequentially solving multiple semi-relaxed MIP sub-problems. This is a useful size-reduction method that requires careful calibration and consideration. Extensive testing was performed for the heuristic design and parameter calibration to ensure good cost-effectiveness in terms of achieving significant computational savings while maintaining the solution quality. The proposed relaxation heuristic is illustrated in Fig. 7.

The algorithm initializes segregating the binary variables from the MIP formulation (often associated with feedstocks, units, tanks, operations, and logistics constraints) into a core group A that includes setup and start-up variables, and a non-core group B that concerns logistic constraints. At each optimization, the non-fixed binary variables from core-group A are relaxed and the model is optimized. From the optimal solution, the binary variables from core-group A within a tolerance δ are fixed to 1 and remain unchanged over future optimizations. While the convergence criteria for δ are not met, the algorithm iteratively updates the tolerance δ and re-optimizes the problem, until meeting any of the following criteria: a) δ reaches its upper bound; or b) all relaxed binaries are fixed. When the convergence criteria for δ are met, the problem is optimized one last time to determine the remaining decision variables in

the problem.

Surrogate modeling: A surrogate model building methodology from Franzoi et al. (2020) is used to build surrogates for the crude distillation unit. The method uses machine learning regression to estimate the yields and properties of distillates using the crude oil assay and the hypothetical improved swing-cuts as input training variables. They are built through measurement feedback by using simulated data. The method is efficient and provides reliable surrogates that can be integrated into scheduling problems with small increase in the computational effort and data requirements. In addition, the surrogates can be continuously updated for enhanced predictions.

Nonlinear programming heuristic: It is well known that starting points for decision variables play a key role in nonlinear optimization problems, especially when there are highly nonlinear and/or nonconvex terms. This affects the effort needed to find optimal solutions and typically leads to poor or infeasible solutions when the starting points are not properly chosen. Because of the complexity of the formulation addressed herein, which might often lead to poor convergence, we developed a heuristic routine for solving the NLP sub-models. Concepts of exploration and exploitation search procedures (see Franzoi et al., 2021b, 2021c) are employed to systematically manage the selection of starting points for the decision variables. The idea consists of an initial wide search across the entire space to identify a potentially good source of solutions, and then subsequently look for better solutions in this particular region.

Within the proposed NLPH routine, three classes of NLP sub-models are defined, in which the difference among them is the selection of starting points. The hyper-parameters that represent the number of instances to be optimized for each class of NLP sub-problem respectively are $\eta_1^{NLP} = 1$, $\eta_2^{NLP} = 3$, and $\eta_3^{NLP} = 3$. The first class considers starting points defined by the user (typically the scheduler). If such information is missing, the algorithm retrieves the last set of optimal starting points used in the previous optimization (if available), or simply skips this step. The solution quality from the NLP optimization strongly relies on how good the starting points are, which represents a potential to quickly achieve fairly good solutions in early stages of the algorithm.

The second class regards the exploration search, in which a combination of fixed and randomized values is employed. The user has flexibility to determine fixed values for any decision variables based on plant requirements, personal knowledge, or other reliable source of information. The starting point $\bar{d}v^{SP}$ for each remaining variable dv is calculated according to Eq. (53), which uniformly lies between the lower and upper bounds $\bar{d}v^L$ and $\bar{d}v^U$ by generating a random number $rand = [0,1]$. A total of η_2^{NLP} optimizations are performed.

$$\bar{d}v^{SP} = rand (\bar{d}v^U - \bar{d}v^L) + \bar{d}v^L \quad (53)$$

The best solution found from the previous optimizations is saved, which comprises the value of each decision variable dv^{opt} . The third class relies on a hybrid idea of exploitation with randomized search, in which the best solution found provides insights for further exploring the optimization space. The starting point of each variable dv is randomly chosen to be as in Eq. (54) but subject to the hard bounds in Eq. (55). The random number $rand = [0,1]$ is multiplied by a hyper-parameter $\delta^{NLP} = \{0.025, 0.05, 0.10\}$ associated with each of the $\eta_3^{NLP} = 3$ optimizations performed for the third class of NLP sub-models.

$$\bar{d}v^{SP} = dv^{opt} (1 + \delta^{NLP} (2 rand - 1)) \quad (54)$$

$$\bar{d}v^L \leq \bar{d}v^{SP} \leq \bar{d}v^U \quad (55)$$

The large degree of nonconvexity typically found in refinery scheduling optimization problems leads to a large number of local optima. This solving procedure provides an efficient fashion to search the neighborhood of the incumbent solution as an alternative to tackle highly nonconvex problems.

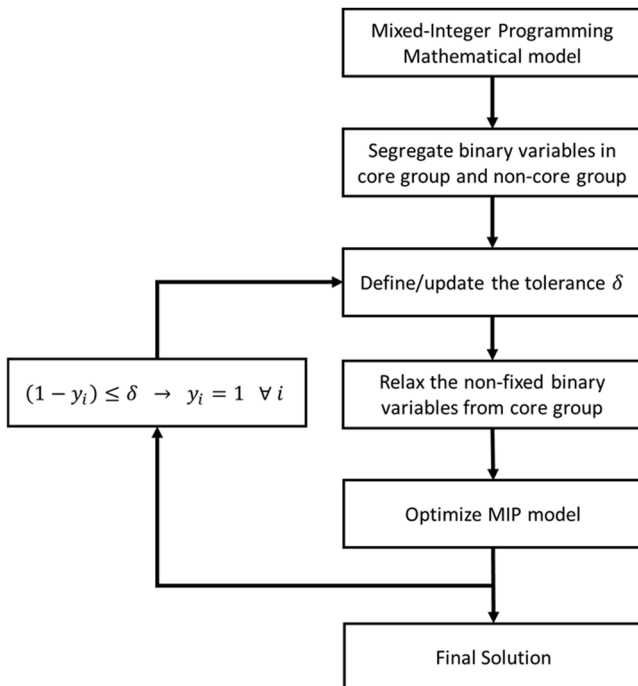


Fig. 7. Algorithm for the proposed relaxation heuristic.

5. Example: large-scale refinery scheduling optimization

The proposed framework is employed to solve a large-scale integrated refinery scheduling optimization problem which comprises the entire refinery value chain, i.e., crude oil blend scheduling, production network, and product blending network. Specifically, the problem topology considers accurate blending and processing networks that are coherent with the physical configuration in the refinery process. This includes the modeling of blending operations with continuous blender units instead of batch mixtures, availability of multiple crude oils that can be blended to produce the distillation feed, the inclusion of tanks prior processing units to provide scheduling flexibility, cascaded distillation network with detailed micro-cut crude oil assay, a complex processing network of unit-operations, and product blending network for production of fuels.

5.1. Problem statement

The refinery network is illustrated in Fig. 2. The crude oil blend scheduling segment includes 12 crude oil feedstocks, 12 storage tanks, 2 blenders, 4 feed tanks, and a distillation network with 5 towers, namely, two atmospheric distillation units, vacuum distillation unit, flash distillation unit, and debutanizer unit. A crude oil assay is used for the calculation of crude oil yields, compositions, and properties, in which a micro-cut distribution (segmented into 10 °C fractions) is considered. Details on the micro-cut calculation and distribution can be found in Menezes et al. (2013). The production network includes fluid catalytic cracking unit, hydrotreaters for coker light naphtha, light cracked naphtha, and diesel, delayed coking unit, two debutanizers, super-fractionator, and catalytic reformer unit. The product blending network includes three blenders for gasoline, diesel, and fuel oil to mix intermediate process streams with boosters for final product specification. A comprehensive explanation of the refinery network, discussing the resources, products, unit-operations, and production process can be found in Supplementary Material C.

The mathematical formulation for the scheduling problem requires information related to:

- Inventory capacity of tanks
- Flowrate capacity of processing units
- Feedstocks (availability and market value)
- Products (demand, market value, and quality specifications)
- Initial inventories and qualities of materials throughout the plant
- Crude oil assay that contains quality information of compositions, yields, and properties over all micro-cut fractions of feedstocks.

The values adopted for such parameters are shown in Table D1 (inventory capacity of tanks), Table D2 (flowrate capacity of processing units), Table D3 (information of feedstocks), Table D4 (information of products), and Table D5 (initial inventories and qualities), presented in Supplementary Material D.

5.2. Computational experiments

The mathematical formulation is built in the commercial modeling and solving platform IMPL (Industrial Modeling & Programming Language) supplied by Industrial Algorithms Limited. The MILP formulation is solved using the commercial optimization solver Gurobi 9.1.1. The NLP formulation is successively linearized through IMPL's sequential linear programming (SLP) algorithm with sub-optimization calls to the Gurobi LP. The use of an SLP algorithm was effective to handle the nonlinearities in a timely fashion; global optimization solvers were not suitable given the size of industrial-scale operations. The machine used is an Intel Core i7 with 2.7 GHz and 16 GB RAM. The total computational time allotted for optimization is 36,000 s.

The framework is tested over a large-scale scheduling optimization problem based on the refinery network in Fig. 2. Multiple scenarios are designed and solved to investigate the computational tractability and solution quality under distinct conditions, i.e., different model sizes, scheduling parameters, and framework hyper-parameters. While larger time horizon length and small time-step size are anticipated to improve profitability and performance, they scale with the model size and therefore should be carefully tuned to provide timely solutions. The scenarios are shown in Table 1 with their respective total time horizon length, segmented time horizon length, and time-step size. The temporal decomposition routine segregates the total time horizon in smaller time

Table 1
Scenarios proposed for the refinery scheduling optimization problem.

Scenario	Time Horizon (days)	Segmented Time horizon (days)	Time-Step (hours)	Objective Function (10 ³ US\$)	Wall-Clock Time (s)
1a	10	10	24	100,079	23
1b	10	5	24	98,351	14
2a	15	15	24	130,969	551
2b	15	5	24	123,707	22
3a	20	20	24	157,786	1346
3b	20	5	24	150,718	30
4a	25	25	24	188,103	2678
4b	25	5	24	175,698	41
5a	30	30	24	218,270	4498
5b	30	5	24	206,143	58
5c	30	3	24	194,222	44
6a	30	30	12	233,760	36,000+
6b	30	5	12	223,060	851
6c	30	3	12	207,905	437
7a	30	30	8	244,615	36,000+
7b	30	5	8	239,152	1806
7c	30	3	8	235,800	730
8a	30	30	4	252,754	36,000+
8b	30	5	4	245,002	3265
8c	30	3	4	241,780	2285
8d	30	1	4	225,488	254
9a	30	30	2	255,899	36,000+
9b	30	5	2	253,990	5108
9c	30	3	2	252,568	3734
9d	30	1	2	234,217	760

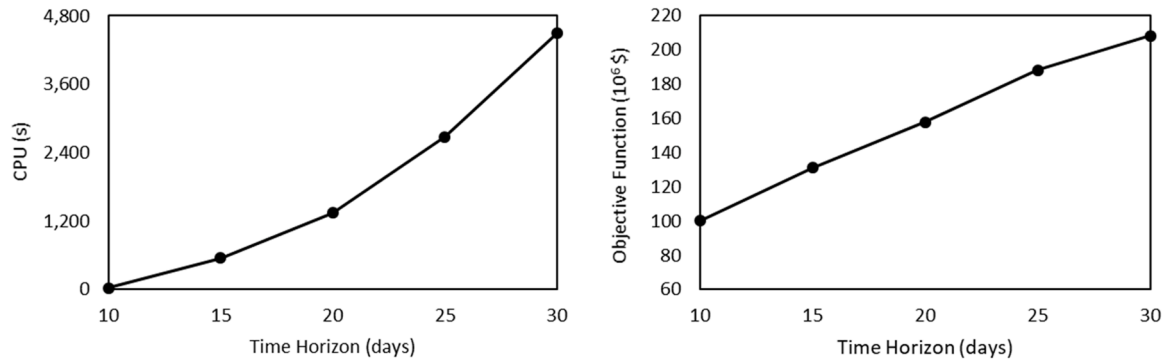


Fig. 8. Solution statistics for the: a) computational time and b) objective function over the variation of the time horizon length.

chunks. The objective function for profit maximization and the wall-clock time are reported as well. Among the scenarios proposed, the largest MINLP formulation considers time horizon of 30 days and time-step of 2 h, for a total of 360 time periods. There are around 70,000 continuous and 50,000 binary variables, 180,000 constraints and 90,000 degrees of freedom as reported by IMPL.

The original MINLP formulation is large in size and highly complex given the high degree of nonlinearities and nonconvexities, mostly from the blending equations. The largest scenario considers hundreds of thousands of variables and constraints in a nonconvex MINLP formulation, which is successfully solved by the proposed framework in reasonable computational time.

Table 1 indicates flexibility regarding the selection of appropriate scheduling parameters according to the computational limitations and requirements towards online scheduling strategies. The selection of scheduling parameters is key for achieving proper trade-off. For example, an optimal solution of \$239 M was found in 30 min using time-steps of 8 h; and an optimal solution of \$252 M was found in 60 min using time-steps of 2 h. In general, a large time horizon with small time-steps provided significant improvements in the solution quality. Scenarios 9a, 9b, and 9c are high-quality solution candidates to be implemented in the plant. It is worth noting that if the total optimization time requires too much effort (e.g., above 2h), online scheduling procedures may not be appropriate. Conversely, enabling enhanced computational resources in the scheduling optimization provides better solutions and significant economic benefits.

Some results from Table 1 are plotted for easier visualization and comparison. Fig. 8 illustrates the computational time and the objective function to be maximized over the increase in the total time horizon length for Scenarios 1a, 2a, 3a, 4a, 5a. Fig. 9 illustrates the computational time (a) and the objective function to be maximized (b) over the decrease in the time-step size for all scenarios with 30-days time horizon. The black line with circular markers represents Scenarios 5a, 6a, 7a, 8a, 9a (no time segmentation); the blue line with triangle markers

represents Scenarios 5b, 6b, 7b, 8b, 9b (time horizon segmentation of 5 days for each sub-problem); and the red line with square markers represents Scenarios 5c, 6c, 7c, 8c, 9c (time horizon segmentation of 3 days for each sub-problem).

The results shown in Table 1 and Figs. 8 and 9 illustrate how the scheduling solution is affected by the proposed methods and the calibration of parameters and hyper-parameters, which is particularly important toward establishing metrics that are relevant for the application and implementation in industrial processes. This includes identifying a proper interplay among the characteristics of the problem to be solved, the scheduling parameters (e.g., time representation, time horizon length, time-step size), and a combination of mathematical optimization techniques. The algorithmic and computational methods comprised in the decision-making framework developed herein enable solving otherwise intractable large-scale scheduling problems, whereby providing reasonably quick solutions to be carried out in a rescheduling fashion.

Franzoi and Menezes (2022) provide some quantitative insights on the impact of scheduling parameters and rescheduling strategies for typical blend scheduling optimization problems. The scheduling horizon is critical for anticipating spot market opportunities and providing more flexible solutions to handle distinct scenarios that may happen in the plant (e.g., demand peaks, maintenance of units). In this context, the selection of large time horizon length (e.g., 30 days) for scheduling applications is especially relevant. The selection of the time-step size is particularly important from an operational perspective. The comparison among Scenarios 5a to 9a, 5b to 9b, and 5c to 9c, indicates significant improvements in the scheduling solution as the time-step size is decreased from 24h to 2h. Small time-steps provide improved decision-making from the additional degrees of freedom in the optimization, which results in better management of resources and production profile.

The importance of employing small time-steps also affects re-optimization features toward achieving online scheduling solutions.

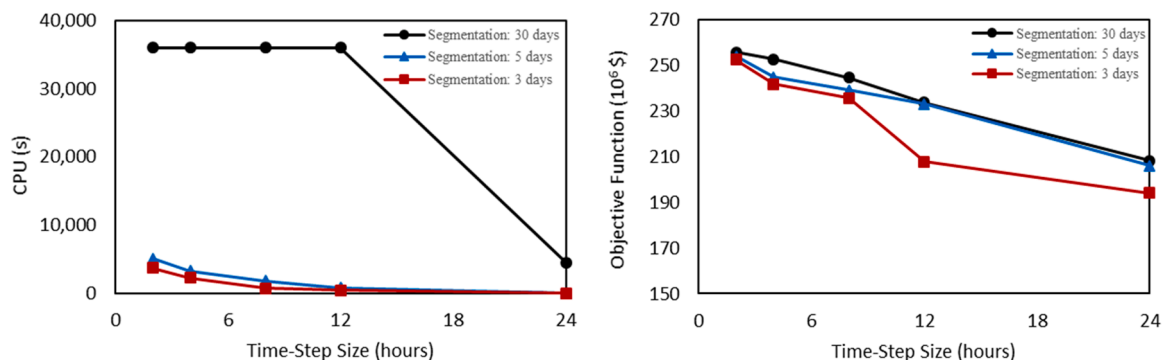


Fig. 9. Solution statistics for the: a) computational time and b) objective function over the variation of the time-step size.

The solution from Scenario 9b (in which a 2h time-step scheduling model is solved in less than 2h) allows addressing online scheduling approaches, whereby the solution is continuously updated within a moving horizon fashion. In such an approach, at certain instants of time (e.g., every 2h or upon event triggering), the incumbent status of the system is assessed, additional information including noises and disturbances and evaluated and added to the model, and the system is re-optimized (similarly to the method developed in Franzoi et al., 2021a). Such insights highlight the usefulness of efficient methodologies to tackle industrial problems whereas providing solution approaches that are implementable and can be further adjusted to meet the industrial needs in terms of operations, constraints, and requirements.

Importantly, the solution approach should be chosen according to the specific needs of the industry or process and the scheduling decision-making core should be coordinated with the plant operations. For example, small time-steps are only useful if the solution can be properly implemented in a timely manner; large time horizon requires reliable and predictable information about the market and operations over the upcoming near future; and online scheduling strategies are leveraged with a sufficient degree of digitized measurement and automation systems. It is also worth highlighting the importance of such features for the long-term investment planning, which concerns multi-level and multi-scope decision-making. The optimization scenarios solved in this work provide meaningful insights associated with the benefits of better scheduling, which is especially helpful in operational and economic analyses toward smarter investment planning, especially lower-cost feedstock selection and the alignment of the product portfolio with market demands. Such considerations are fundamental to further enhance the performance of industrial processes.

6. Main remarks and contributions of this work

There has been recent research and technological advancements in computer power, solution algorithms, and decision-making with applications in computer-aided and process systems engineering. Toward further breakthroughs in these field, we address the modeling and solving of industrial-scale refinery scheduling optimization applications. The main contributions of this work are threefold. First, we discuss the state-of-the-art limitations of industrial-scale refinery scheduling and we provide an overview of efficient modeling and algorithmic methods for tackling this problem. Second, we present a novel mathematical model that accurately represents refinery scheduling operations considering a full-scope refinery topology and relevant operational constraints. Third, we propose a novel decision-making framework based on mathematical modeling and optimization capabilities to solve industrial-scale non-convex MINLP refinery scheduling systems. To the best of our knowledge, this is the first work that proposes tackling and solving refinery scheduling optimization problems in such a scope, size, scale and complexity.

The framework leverages the use of mathematical optimization and algorithmic methods by combining modeling approaches (process design, model decompositions), solving strategies (rescheduling procedures, heuristic algorithms), and machine learning regression (surrogate models). It successfully solves an industrial-size refinery scheduling optimization problem formulated as a nonconvex mixed-integer nonlinear programming (MINLP) model, providing more efficient scheduling operations and higher profitability and performance. The formulation is coherent with industrial applications in terms of its operational constraints and processing complexity. From the results presented herein, we anticipate promising research on the development of solution methods that enhance industrial decision-making in the process systems engineering (PSE) field.

6.1. Future outlook on industrial-scale refinery scheduling optimization

We highlight six features that should be considered toward the future

generation of scheduling optimization technology for improved refinery decision-making. From a modeling perspective, this includes the consideration of an integrated refinery scheduling network, realistic mathematical formulation based on plant requirements, and accurate process-unit predictions. From a solving perspective, this includes optimization decision-making, consideration of scheduling parameters, and re-optimization mechanisms.

1. **Integrated Network:** Simultaneously solving an integrated problem including crude oil scheduling, processing network, and product blending network in the same formulation, exploits additional degrees of freedom and leads to improved decision-making.
2. **Realistic Formulation:** The scheduling formulation needs to be coherent with the operational requirements and limitations in the plant. Realistic models lead to realistic schedules that are more easily and smoothly implemented.
3. **Accurate Predictions:** Process unit models need to be represented in an accurate yet tractable fashion. Surrogate modeling plays an important role in the integration between process unit models and refinery scheduling optimization.
4. **Optimization Decision-making:** Refinery scheduling has been transitioning from trial-and-error to simulation-aided and more recently to optimization-aided decision-making. Optimization methods are key to manage complex operations and maximize profitability and performance.
5. **Scheduling Parameters:** Tuning scheduling parameters (e.g., time-step, time horizon, rescheduling frequency) provides better solution and easier implementation in the plant. Trade-offs between solution quality and computational tractability must be assessed.
6. **Rescheduling:** Mechanisms to re-optimize and automate the scheduling decision-making provide efficient and timely capabilities for managing such complex and uncertain process.

The development of advanced computer-aided optimization decision-making is fundamental for further advances in the process systems engineering field, and the abovementioned features are key toward state-of-the-art improvements in refinery scheduling decision-making. An efficient decision-making framework that comprises accurate formulations and computer-aided solving algorithms enables improved solutions for refinery scheduling, with higher efficiency and lower costs. Proper assessment of scheduling parameters and tuning the framework hyper-parameters provides a proper trade-off between solution quality and computational effort that enables its application for industrial operations.

CRediT authorship contribution statement

Robert E. Franzoi: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Conceptualization. **Brenno C. Menezes:** Methodology, Conceptualization. **Jeffrey D. Kelly:** Methodology, Writing – review & editing, Software. **Jorge A.W. Gut:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Ignacio E. Grossmann:** Writing – review & editing, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

The authors would like to acknowledge the support from: FAPESP - Fundação de Amparo à Pesquisa do Estado de São Paulo (grants 2017/03310-1 and 2018/04942-4), CAPES - Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (Finance Code 001), and CNPq - Conselho Nacional de Desenvolvimento Científico e Tecnológico (grant 316388/2021-1).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.compchemeng.2024.108678.

References

- Ali, T.H.M., Franzoi, R.E., Menezes, B.C., 2022. Surrogate modeling for nonlinear gasoline blending operations. *Comput. Aided Chem. Eng.* 49, 1783–1788. Elsevier.
- Al-Qahtani, K.Y., Elkamel, A., 2011. Planning and Integration of Refinery and Petrochemical Operations. John Wiley & Sons.
- Brunaud, B., Perez, H.D., Amaran, S., Bury, S., Wassick, J., Grossmann, I.E., 2020. Batch scheduling with quality-based changeovers. *Comput. Chem. Eng.* 132, 106617.
- Castillo Castillo, P., Castro, P.M., Mahalec, V., 2017. Global optimization algorithm for large-scale refinery planning models with bilinear terms. *Ind. Eng. Chem. Res.* 56 (2), 530–548.
- Castro, P.M., Grossmann, I.E., 2014. Global optimal scheduling of crude oil blending operations with RTN continuous-time and multiparametric disaggregation. *Ind. Eng. Chem. Res.* 53 (39), 15127–15145.
- Castro, P.M., Grossmann, I.E., Zhang, Q., 2018. Expanding scope and computational challenges in process scheduling. *Comput. Chem. Eng.* 114, 14–42.
- Chrysosolouris, G., Papakostas, N., Mourtzis, D., 2005. Refinery short-term scheduling with tank farm, inventory and distillation management: an integrated simulation-based approach. *Eur. J. Oper. Res.* 166 (3), 812–827.
- Cuadros Bohorquez, J.F., Plazas Tovar, L., Wolf Maciel, M.R., Melo, D.C., Maciel Filho, R., 2020. Surrogate-model-based, particle swarm optimization, and genetic algorithm techniques applied to the multiobjective operational problem of the fluid catalytic cracking process. *Chem. Eng. Commun.* 207 (5), 612–631.
- Franzoi, R.E., Ali, T., Al-Hammadi, A., Menezes, B.C., 2021c. Surrogate modeling approach for nonlinear blending processes. In: 2021 1st International Conference on Emerging Smart Technologies and Applications. IEEE, pp. 1–8.
- Franzoi, R.E., Kelly, J.D., Menezes, B.C., Swartz, C.L.E., 2021b. An adaptive sampling surrogate model building framework for the optimization of reaction systems. *Comput. Chem. Eng.* 152, 107371.
- Franzoi, R.E., Menezes, B.C., 2022. Towards improved scheduling: an analysis on time-steps, time horizon, and rescheduling. *Comput. Aided Chem. Eng.* 51, 1003–1008. Elsevier.
- Franzoi, R.E., Menezes, B.C., Kelly, J.D., Gut, J.A., 2021a. A moving horizon rescheduling framework for continuous nonlinear processes with disturbances. *Chem. Eng. Res. Des.* 174, 276–293.
- Franzoi, R.E., Menezes, B.C., Kelly, J.D., Gut, J.W., 2018a. Effective scheduling of complex process-shops using online parameter feedback in crude oil refineries. *Comput. Aided Chem. Eng.* 44, 1279–1284. Elsevier.
- Franzoi, R.E., Menezes, B.C., Kelly, J.D., Gut, J.A., 2019. Design for online process and blend scheduling optimization. *Comput. Aided Chem. Eng.* 47, 187–192. Elsevier.
- Franzoi, R.E., Menezes, B.C., Kelly, J.D., Gut, J.A., Grossmann, I.E., 2020. Cutpoint temperature surrogate modeling for distillation yields and properties. *Ind. Eng. Chem. Res.* 59 (41), 18616–18628.
- Garcia, D.J., Jian, G., Fengqi, Y., 2016. Multi-stage adaptive robust optimization over bioconversion product and process networks with uncertain feedstock price and biofuel demand. *Comput. Aided Chem. Eng.* 38, 217–222. Elsevier.
- Kelly, J.D., 2002. Chronological decomposition heuristic for scheduling: divide and conquer method. *AIChE J.* 48 (12), 2995–2999.
- Kelly, J.D., 2005. The unit-operation-stock superstructure (UOSS) and the quantity-quality paradigm (QLQP) for production scheduling in the process industries. In: MISTA 2005 Conference Proceedings, Stern School of Business. New York University.
- Kelly, J.D., Mann, J.L., 2003. Crude-oil blend scheduling optimization: an application with multi-million dollar benefits. *Hydrocarbon Processing*, June/July.
- Kelly, J.D., Zyngier, D., 2007. An improved MILP modeling of sequence-dependent switch-overs for discrete-time scheduling problems. *Ind. Eng. Chem. Res.* 46.
- Kelly, J.D., Zyngier, D., 2017. Unit-operation nonlinear modeling for planning and scheduling applications. *Optim. Eng.* 18, 133–154.
- Kelly, J.D., Menezes, B.C., Engineer, F., Grossmann, I.E., 2017. Crude oil blend scheduling optimization of an industrial-sized refinery: a discrete-time benchmark. *Foundations of Computer Aided Process Operations*. Tucson, United States.
- Kelly, J.D., Menezes, B.C., Grossmann, I.E., 2018. Successive LP approximation for nonconvex blending in MILP scheduling optimization using factors for qualities in the process industry. *Ind. Eng. Chem. Res.* 57 (32), 11076–11093.
- Lee, H., Pinto, J.M., Grossmann, I.E., Park, S., 1996. Mixed-integer linear programming model for refinery short-term scheduling of crude oil unloading with inventory management. *Ind. Eng. Chem. Res.* 35 (5), 1630–1641.
- Li, F., Yang, M., Du, W., Dai, X., 2020. Development and challenges of planning and scheduling for petroleum and petrochemical production. *Front. Eng. Manag.* 1–11.
- Mencarelli, L., Pagot, A., Duchêne, P., 2020. Surrogate-based modeling techniques with application to catalytic reforming and isomerization processes. *Comput. Chem. Eng.* 135, 106772.
- Menezes, B.C., Kelly, J.D., Grossmann, I.E., 2013. Improved swing-cut modeling for planning and scheduling of oil-refinery distillation units. *Ind. Eng. Chem. Res.* 52 (51), 18324–18333.
- Menezes, B.C., Kelly, J.D., Grossmann, I.E., 2015. Phenomenological decomposition heuristic for process design synthesis of oil-refinery Units. *Comput. Aided Chem. Eng.* 37, 1877–1882. Elsevier.
- Mouret, S., Grossmann, I.E., Pestiaux, P., 2009. A novel priority-slot based continuous-time formulation for crude oil scheduling problems. *Ind. Eng. Chem. Res.* 48 (18), 8515–8528.
- Rafiei, M., Ricardez-Sandoval, L.A., 2020. New frontiers, challenges, and opportunities in integration of design and control for enterprise-wide sustainability. *Comp. Chem. Eng.* 132, 106610.
- Saharidis, G.K.D., Minoux, M., Dallery, Y., 2009. Scheduling of loading and unloading of crude oil in a refinery using event-based discrete time formulation. *Comput. Chem. Eng.* 33 (8), 1413–1426.
- Shah, N.K., Sahay, N., Ierapetritou, M.G., 2015. Efficient decomposition approach for large-scale refinery scheduling. *Ind. Eng. Chem. Res.* 54 (41), 9964–9991.
- Song, W., Du, W., Fan, C., Yang, M., Qian, F., 2021. Adaptive weighted hybrid modeling of hydrocracking process and its operational optimization. *Ind. Eng. Chem. Res.* 60 (9), 3617–3632.
- Uribe-Rodríguez, A., Castro, P.M., Gonzalo, G.G., Chachuat, B., 2020. Global optimization of large-scale MIQCQPs via cluster decomposition: application to short-term planning of an integrated refinery-petrochemical complex. *Comput. Chem. Eng.* 140, 106883.
- Uribe-Rodríguez, A., Castro, P.M., Guillén-Gosálbez, G., Chachuat, B., 2023. Assessment of Lagrangean decomposition for short-term planning of integrated refinery-petrochemical operations. *Comput. Chem. Eng.* 174, 108229.
- Winston, W.L., Goldberg, J.B., 2004. *Operations Research: Applications and Algorithms*. Thomson Brooks/Cole, Belmont. Vol. 3.
- Xia, Z., Wang, S., Zhou, L., Dai, Y., Dang, Y., Ji, X., 2021. Surrogate-assisted optimization of refinery hydrogen networks with hydrogen sulfide removal. *J. Clean. Prod.* 310, 127477.
- Xu, J., Zhang, S., Zhang, J., Wang, S., Xu, Q., 2017. Simultaneous scheduling of front-end crude transfer and refinery processing. *Comput. Chem. Eng.* 96, 212–236.
- Yang, H., Bernal, D.E., Franzoi, R.E., Engineer, F.G., Kwon, K., Lee, S., Grossmann, I.E., 2020. Integration of crude-oil scheduling and refinery planning by Lagrangean Decomposition. *Comput. Chem. Eng.* 138, 106812.
- Zhang, J., Rong, G., Hou, W., Huang, C., 2010. Simulation based approach for optimal scheduling of fuel gas system in refinery. *Chem. Eng. Res. Des.* 88 (1), 87–99.
- Zyngier, D., Kelly, J.D., 2009. Multi-product inventory logistics modeling in the process industries. *Optimization and Logistics Challenges in the Enterprise*. Springer, Boston, MA, pp. 61–95.
- Zyngier, D., Kelly, J.D., 2012. UOPSS: a new paradigm for modeling production planning & scheduling systems. In: ESCAPE 22 Conference Proceedings.
- Zyngier, D., Lategan, J., Furstenberg, L., 2018. A process systems approach for detailed rail planning and scheduling applications. *Comput. Chem. Eng.* 114, 273–280.